Core Finance
Week 7

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

RICE | BUSINESS
Jones Graduate School of Business

Week 7

- Efficient Markets Hypothesis
- Behavioral Finance

Beta Management Case Was Done Well

- Suppose we want to calculate betas for two stocks (A & B). Some of you did this regression:

$$r_{M,t} = \alpha + \beta_A(r_{A,t} - r_{rf,t}) + \beta_B(r_{B,t} - r_{rf,t}) + \varepsilon_t$$

This is incorrect.

RICE | BUSINESS
Jones Graduate School of Business

Beta Management Case Was Done Well

- Suppose we want to calculate betas for two stocks (A & B). Some of you did this regression:

$$r_{M,t} = \alpha + \beta_A\left(r_{A,t} - r_{rf,t}\right) + \beta_B\left(r_{B,t} - r_{rf,t}\right) + \varepsilon_t$$

This is incorrect.

- You need to have the **individual stock on the left-hand-side**. This is because the individual stock returns are the outcome (dependent variable) that we are trying to explain.
- You need to calculate betas using **separate regressions** for stock A and stock B. This is because if stock A and stock B are correlated, it will bias the estimates.

Correct approach:

$$r_{A,t} = \alpha + \beta_A\left(r_{M,t} - r_{rf,t}\right) + \varepsilon_t$$

$$r_{B,t} = \alpha + \beta_B\left(r_{M,t} - r_{rf,t}\right) + \varepsilon_t$$

Thanks for Survey Feedback

Why Do I Use Cold Calling?

- Helps me to work out what people haven't understood

Thanks for Survey Feedback

Why Do I Use Cold Calling?

- Helps me to work out what people haven't understood

Demand for more cases:

- **Applied Finance (648) elective course does this**: practical applications with cases/excel. Highly recommend you take this!

Thanks for Survey Feedback

Why Do I Use Cold Calling?

- Helps me to work out what people haven't understood

Demand for more cases:

- **Applied Finance (648) elective course does this**: practical applications with cases/excel. Highly recommend you take this!

More real-life examples

Some say not enough problems, other say too many

Some say not enough step-by-step / too fast, others say too much / too slow

First Half ✅ Second Half

Week	Topics	After Class Assignments
7 (31 March, 2 April)	Efficient Markets, Behavioral Finance	
8 (7, 9 April)	Financing and Capital Structure, MM Theory	Quiz 6 Case - Boeing 7E7
9 (14, 16 April)	Boeing 7E7 (Case), Discounted & Free Cash Flows	Quiz 7
10 (21, 23 April)	Options	Quiz 8
11 (28, 30 April)	Great Recession, Consumer Finance	Quiz 9
12 (5, 7 May)	Miscellaneous Topics, Course Review	Final (3 hour canvas exam, 1 week to take)

RICE | BUSINESS
Jones Graduate School of Business

Efficient Markets

Jargon Busting Investment Firms

Asset Managers: Firms that manage investments on behalf of clients. Largest are Blackrock, Vanguard, Fidelity, and State Street Global Advisors.

Mutual Funds: Pools investments from multiple investors. Does not trade on an exchange so can only buy/sell mutual fund once per day based on NAV. Firms typically "**active management**" meaning the "**Fund Manger**" is picking stocks to buy and sell.

Exchange Traded Funds (ETFs): Pools investments from multiple investors. Trades on an exchange and so can buy/sell ETF intra-day based on NAV. "Passive Management" – funds are just following an index (e.g., S&P500).

Net Asset Value (NAV) is per-share market value = (fund's assets – fund's liabilities)/ number of shares. N.b. this is cost to buy/sell but should judge a fund's performance by its total return.

Hedge Funds: Pools investments from multiple investors and implements active investments using complex and riskier strategies (leverage, short-selling)

Private Equity (PE) Firms: Investing in private companies (i.e., not listed on public exchange)

Venture Capital (VC) Funds: Private equity focused on start-ups

Buy-Side: Firms that buy securities to invest (e.g., asset managers)

Sell-Side: Firms that facilitate trades (e.g., investment banks)

Can You Beat the Market?

A Nobel-Prize Winning Lecture!

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2013

© Nobel Media AB. Photo: A. Mahmoud
Eugene F. Fama
Prize share: 1/3

© Nobel Media AB. Photo: A. Mahmoud
Lars Peter Hansen
Prize share: 1/3

© Nobel Media AB. Photo: A. Mahmoud
Robert J. Shiller
Prize share: 1/3

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2017

© Nobel Media AB. Photo: A. Mahmoud
Richard H. Thaler
Prize share: 1/1

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2002

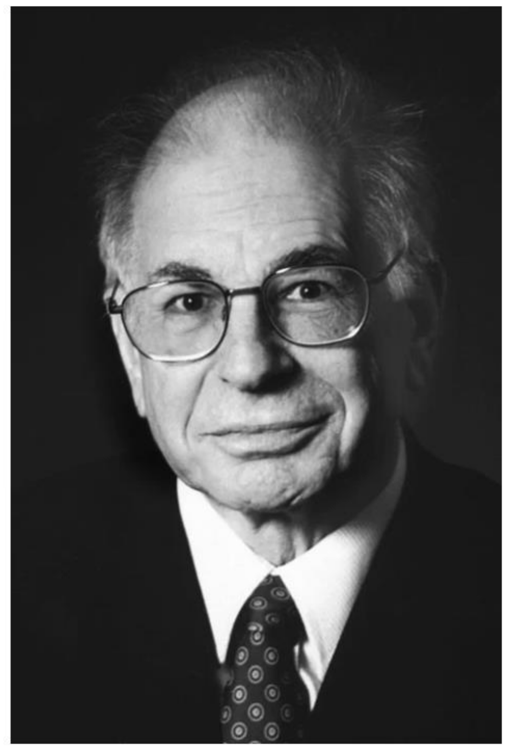


Photo from the Nobel Foundation archive.
Daniel Kahneman
Prize share: 1/2

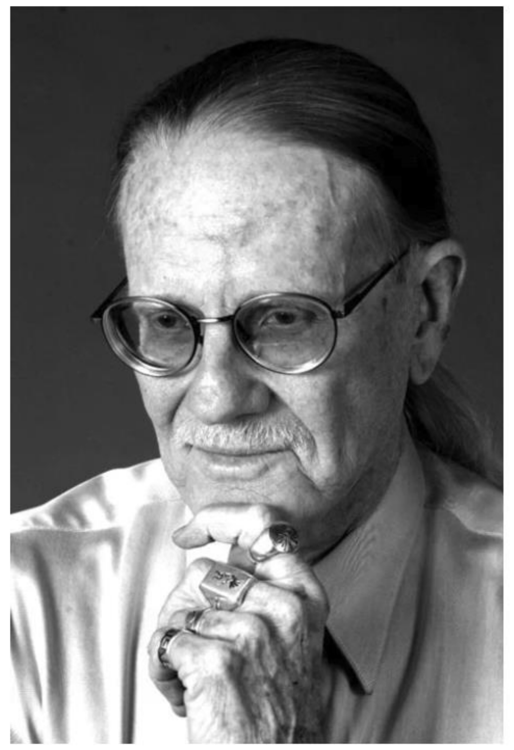


Photo from the Nobel Foundation archive.
Vernon L. Smith
Prize share: 1/2

Investment vs. Financing

Investment Opportunities

- Focus of course this far, e.g., given cashflows, should we invest?

Investment vs. Financing

Investment Opportunities

- Focus of course this far, e.g., given cashflows, should we invest?

Financing Opportunities

- How to finance such investments?
- Markets looking for 'profitable' opportunities

$$\boldsymbol{Price} = \sum_{t=0}^{\infty} \frac{\boldsymbol{C_t}}{(\mathbf{1}+\boldsymbol{r})^{\boldsymbol{t}}}$$

RICE | BUSINESS
Jones Graduate School of Business

Relating Back to PV

$$\boldsymbol{Price} = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

Evaluate whether the ***price*** of assets reflect all information on the right hand side

I.e., price reflects fundamentals (cashflows) and compensating for systematic risk

- If and how fast does **information** get reflected into prices?
- Do markets get this **price right**?
- Can investors find **undervalued/overvalued** assets in the market?

RICE | BUSINESS
Jones Graduate School of Business

Efficient Market Hypothesis

RICE | BUSINESS
Jones Graduate School of Business

Prior to 1960s, finance had little to say about investing approaches

Common investment practices:

“Technical Analysis” / “Charting” – analyze past price/return patterns.

“Fundamental Analysis” – analyze financial/economic data.

Efficient Market Hypothesis (EMH)

Eugene Fama (1970) “The Father of Modern Finance” developed the efficient market hypothesis.

Efficient Market Hypothesis (EMH)

Eugene Fama (1970) “The Father of Modern Finance” developed the efficient market hypothesis.

Markets are “efficient” if prices reflect *all* available information

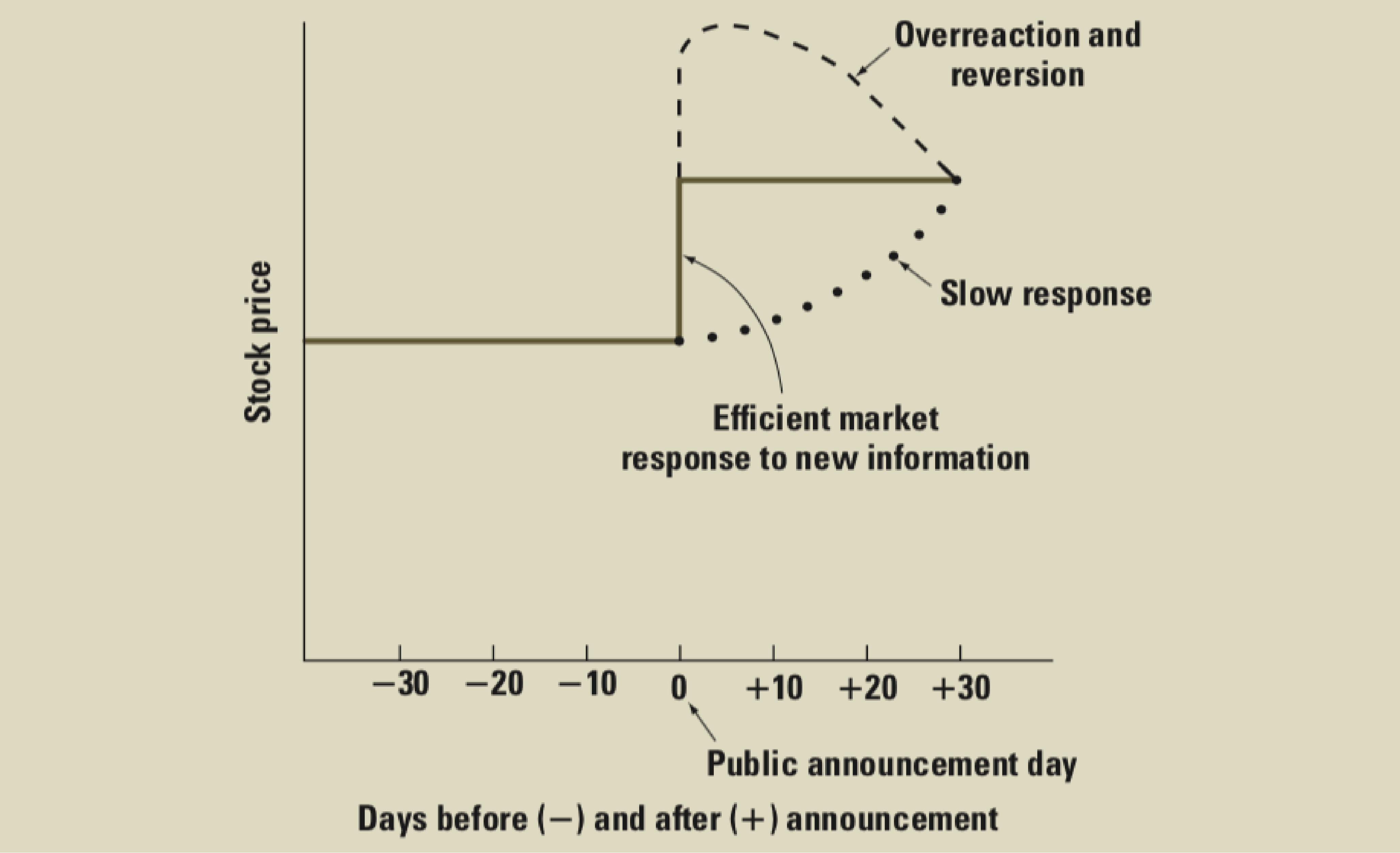
- **Reflect fundamentals of valuation (cash flows)**
- **Appropriate compensation for systematic risk**

Efficient Market Hypothesis (EMH)

Markets are "efficient" if prices reflect *all* available information

Information is immediately reflected in prices

- **Prices adjust before investor has time to trade on new information**



Efficient Market Hypothesis (EMH)

Markets are "efficient" if prices reflect *all* available information

Information is immediately reflected in prices

- Prices adjust before investor has time to trade on new information

Firms should expect to receive "fair value" for selling securities

- Fair value means that the security's price is the present value

Guess How Many Candy Sweets in the Jar
Closest Wins The Jar

https://forms.gle/SUXWiBgj7GeFL33j8

Example of how information -> asset prices

Investors judging value of a firm

Vote with dollars

Investors can have different beliefs

Repeat over time

What EMH does ***not*** imply

- Just randomly choose assets

- NO! EMH does not imply this! You can create portfolios to diversify risk. And then which portfolio to choose depends on your risk preferences.

What EMH does ***not*** imply

- **Just randomly choose assets**

- NO! EMH does not imply this! You can create portfolios to diversify risk. And then which portfolio to choose depends on your risk preferences.

- **Stock prices should not fluctuate**

- NO! EMH does not imply this. New information always being reflected in prices

What EMH does ***not*** imply

- Just randomly choose assets

- NO! EMH does not imply this! You can create portfolios to diversify risk. And then which portfolio to choose depends on your risk preferences.

- Stock prices should not fluctuate

- NO! EMH does not imply this. New information always being reflected in prices

- Need a lot of investors for efficiency

- NO! EMH does not imply this.

How Can Markets Be Efficient?

Prices are the market clearing price based on **supply** and **demand**

Asset prices move **but supply is fixed in short-run** (e.g. number of outstanding shares), so **short-run price changes** must be due to **changes in investor demand**

Demand changes based on information. 3 Different ways to get to this:

1. Everyone is rational, agrees on how to interpret new information & so agrees on P

2. Some people disagree. “Irrational noise traders” vs. rational (or alternatively optimists and pessimists). The two cancel out (by luck?!)

3. “Arbitrage”. Really deep-pocketed arbitrageurs can keep price at the efficient level. For example, always willing to buy when price is below valuation.

Versions of Market Efficiency in EMH

1. Weak Form Efficiency

Prices fully reflect **past price or return information**

- No investor can achieve excess returns by trading based on historical information (e.g., “Technical Analysis” of past price movements)

Versions of Market Efficiency in EMH

1. Weak Form Efficiency

Prices fully reflect **past price or return information**

- No investor can achieve excess returns by trading based on historical information (e.g., “Technical Analysis” of past price movements)

2. Semi-Strong Form Efficiency

Prices fully reflect **all publicly available information (past and present)**

- Public information includes annual reports of corporations, newspapers, investors’ newsletters, etc.

- No investor can achieve excess returns by trading on public information (e.g., “Fundamental analysis” of businesses)

Versions of Market Efficiency in EMH

1. **Weak Form Efficiency**

Prices fully reflect **past price or return information**

- No investor can achieve excess returns by trading based on historical information (e.g., "Technical Analysis" of past price movements)

2. Semi-Strong Form Efficiency

Prices fully reflect **all publicly available information (past and present)**

- Public information includes annual reports of corporations, newspapers, investors' newsletters, etc.

- No investor can achieve excess returns by trading on public information (e.g., "Fundamental analysis" of businesses)

3. Strong Form Efficiency

Prices fully reflect **all publicly *and* all privately available information**

- No investor can earn excess returns upon any type of information, including insider information
- If so, laws against "insider trading" (illegally trading on confidential information) are not necessary

Testing Efficient Market Hypothesis

Joint Hypothesis Problem

A test of market efficiency can only be conducted using a theoretical model to define "fair prices" / "normal returns"

How to interpret "abnormal/excess returns" where prices $\neq$ theoretical model?

- Reject that markets are efficient, do not reject theoretical model
- Reject theoretical model of normal returns, do not reject market efficiency
- Reject both theoretical model and market efficiency

Can't separately test market efficiency.

Testing **Weak Form** Efficiency

"Random Walk Hypothesis" that stock market price in the past is unrelated to its price movement in the future

- Test using "serial correlation" test

$$R_t = \alpha + \gamma R_{t-1} + \varepsilon_t$$

- Is $\gamma = 0$?

Testing **Weak Form** Efficiency

"Random Walk Hypothesis" that stock market price in the past is unrelated to its price movement in the future

- Test using "serial correlation" test

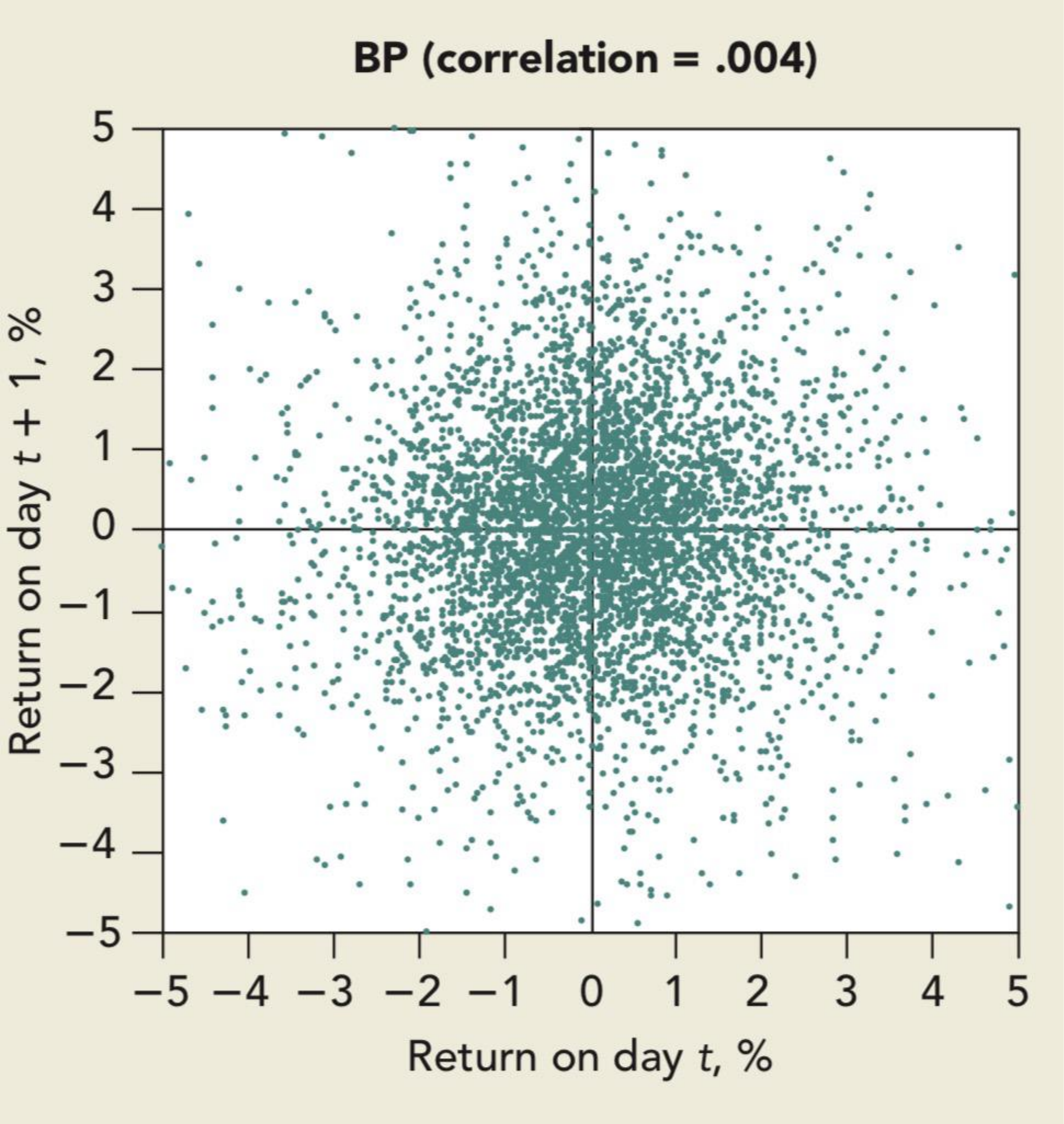
$$R_t = \alpha + \gamma R_{t-1} + \varepsilon_t$$

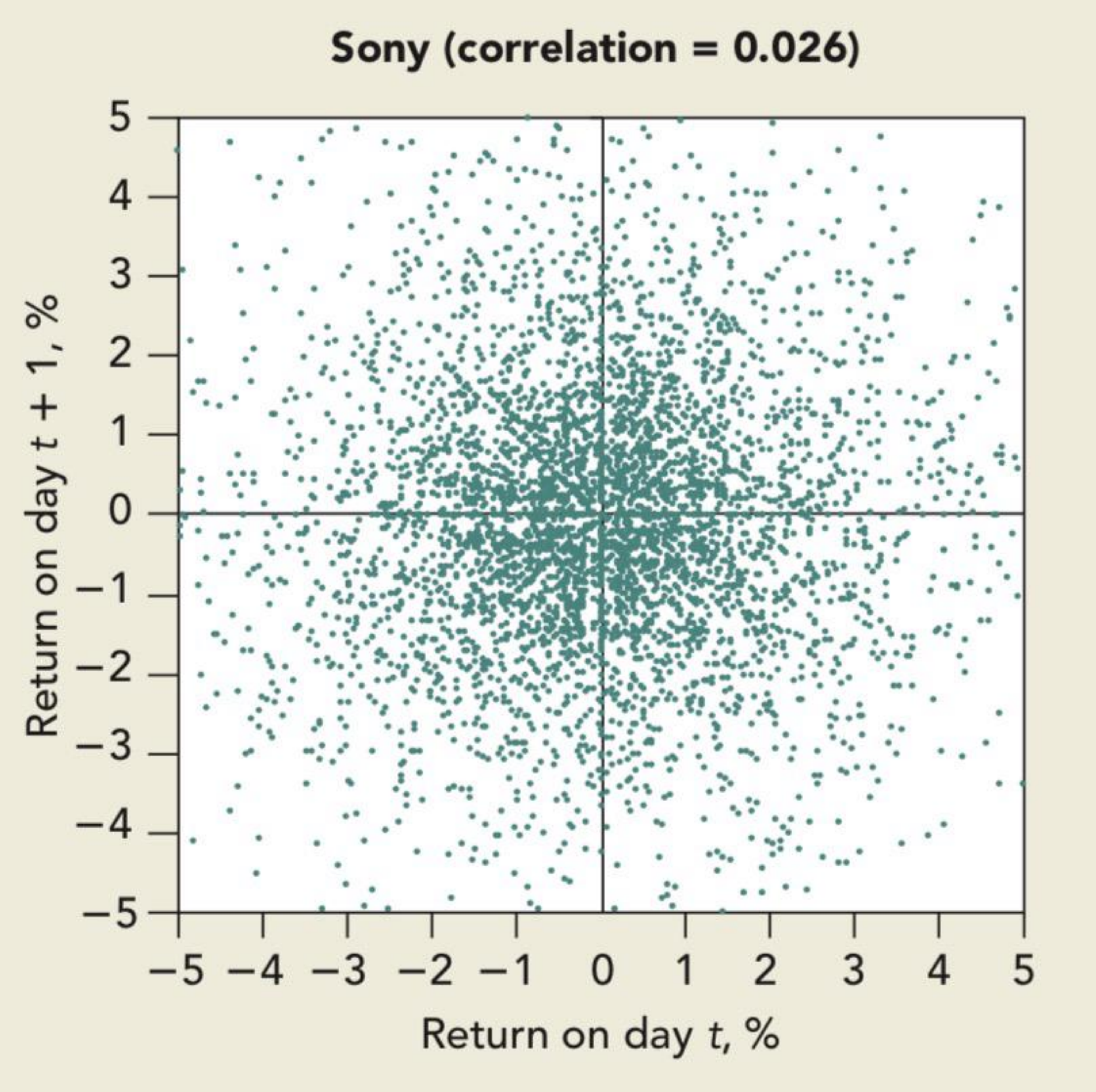
- **Is $\gamma = 0$?**
- **No, but almost.**
- **Very small positive serial correlation, and no profits to make after transaction costs.**

Under weak form efficiency, no investor can achieve excess returns by trading based on historical information (e.g., "Technical Analysis" of past price movements)

- **Evidence shows that technical analysis doesn't yield significant excess returns (except for momentum strategies).**

Visual Examples of Random Walk





Testing **Strong Form** Efficiency

Prices fully reflect **all publicly *and* all privately available information**

Clearly false. Insider trading earns excess profits. Don't do it. It is illegal. SEC is very sophisticated at catching!

Testing **Semi-Strong Form** Efficiency

Look for deviations in prices, based on public information.

Cottage industry of papers finding examples violating this.

After publication of these papers, markets adapt, and the effects weaken or completely go away (McLean, 2016)

Examples to follow later in class.

Challenger Disaster

- 11:39 ET on 28 January 1986 Challenger space shuttle tragically exploded
- 4 manufacturers for shuttle: Lockheed, Martin Marietta, Rockwell, Morton-Thiokol
- Panel investigated cause of explosion and concluded on 6 June 1986
- Cause was defective seals supplied by Morton Thiokol

What do you think happened to the stock price of the four manufacturers?

Challenger Disaster (Maloney & Mulherin, 2003)

- 11:39 ET on 28 January 1986 Challenger space shuttle tragically exploded
- 4 manufacturers: Lockheed, Martin Marietta, Rockwell, Morton-Thiokol
- Panel investigated cause of explosion and concluded on 6 June 1986
- Cause was defective seals supplied by Morton Thiokol
- **Market had already worked this out on 28 January 1986 (prices adjusted)**

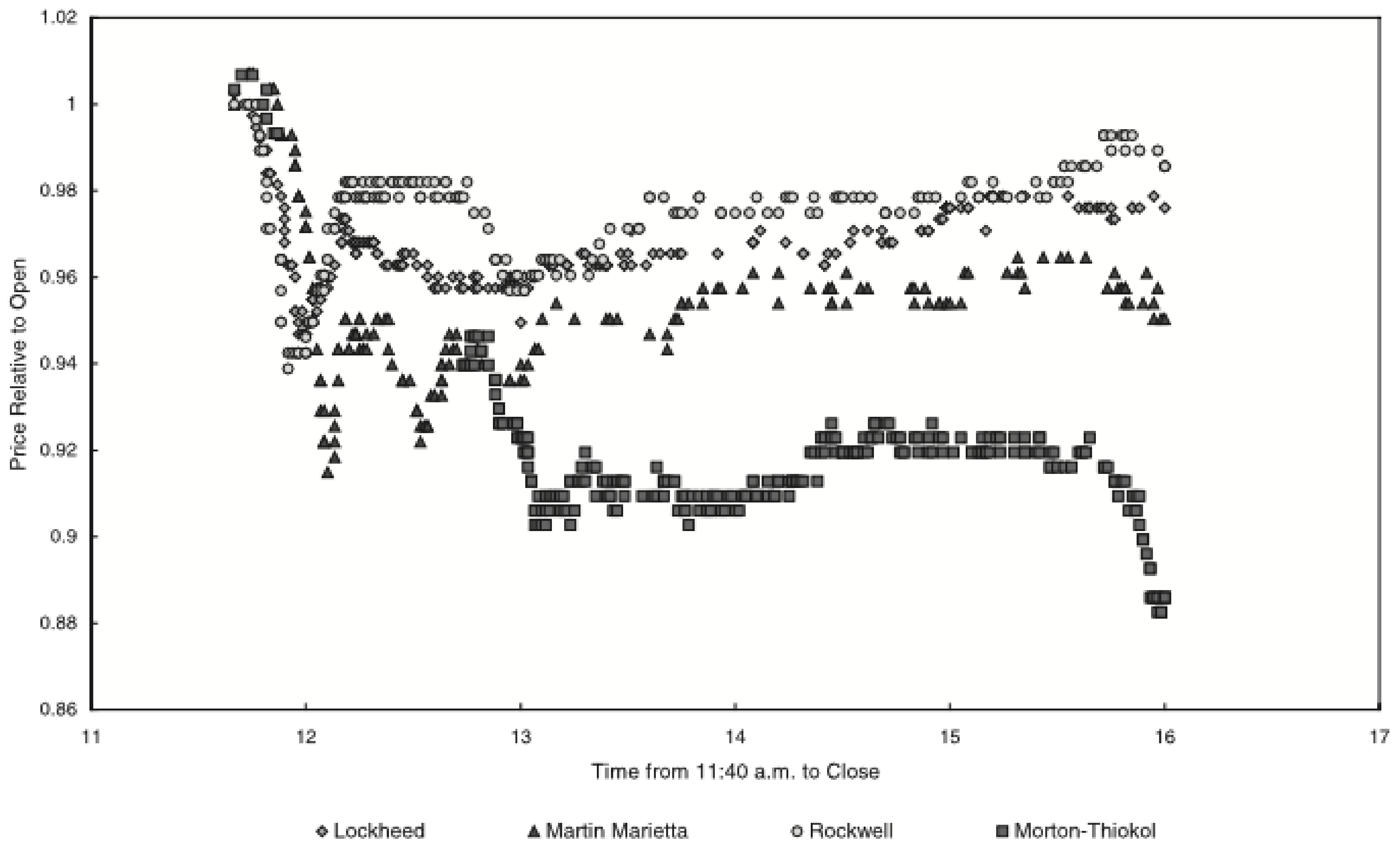


Fig. 1. Intraday stock price movements following the challenger disaster.

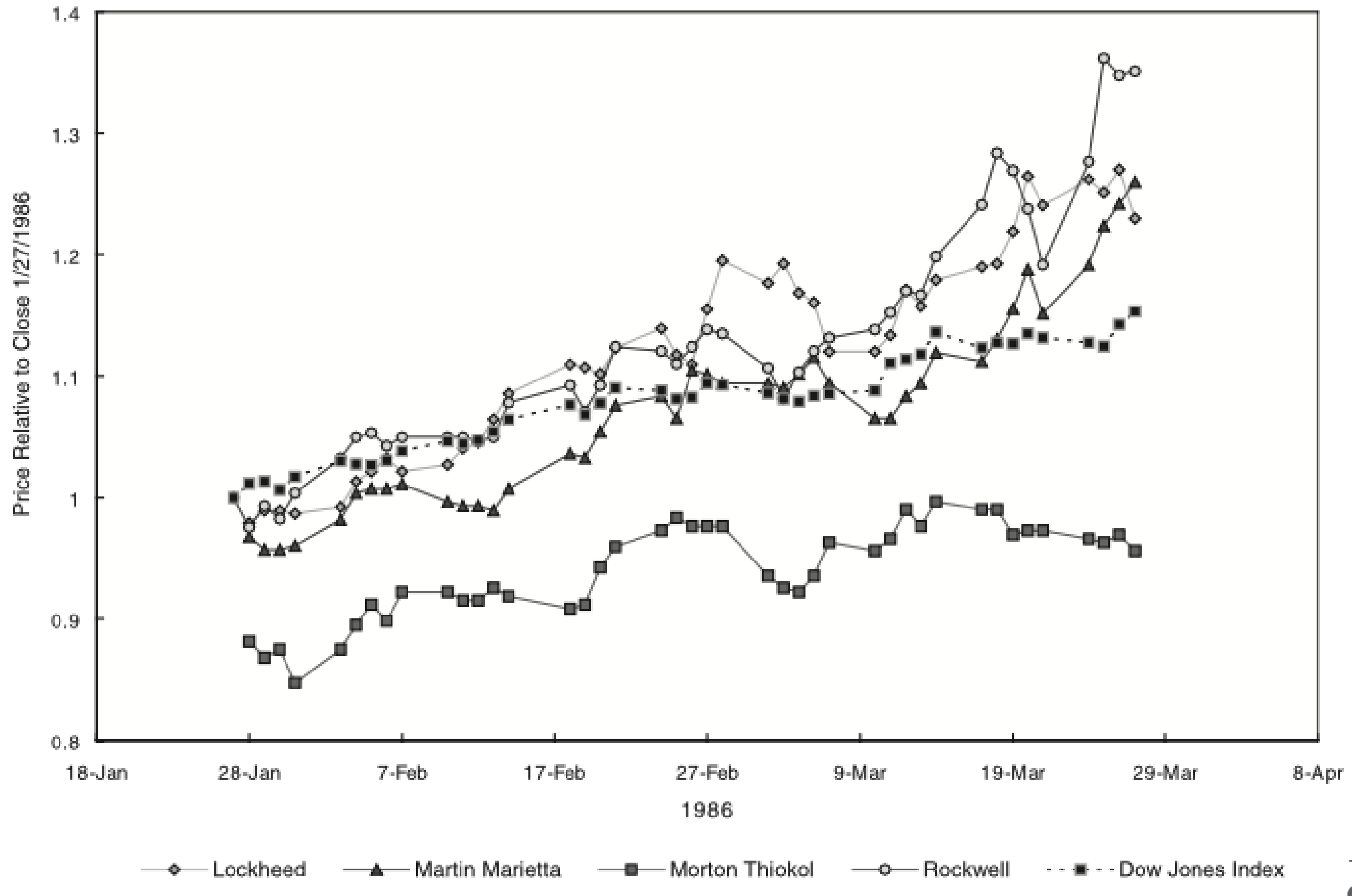


Fig. 2. Three-month stock price movements following the challenger disaster.

NESS of Business

Active or Passive Investing

RICE | BUSINESS
Jones Graduate School of Business

Week 5 Reminder: Implications of Asset Pricing Models for Investments

If CAPM is correct…

You ***cannot*** beat the market, on average.

Instead, passively invest in diversified index fund, minimize your fees.

If CAPM is incorrect…

You ***can*** beat the market, on average.

Make active investments that exploit predictable mis-pricing.

Irrespective of CAPM being correct or incorrect…

Individual investment choices will differ with risk tolerances.

Should You Just Hire an Asset Manager?

RICE | BUSINESS
Jones Graduate School of Business

Jensen (1968) and many others since

- Studied 115 mutual funds and found, on average, they couldn't outperform buy and hold strategy.
- Any fund could not do better than expected by random chance.
- Returns don't justify their fees.

Very robust finding. Mutual funds can outperform the market one year, but rarely repeatedly. Their higher fees outweigh the gains.

Buffet’s $1m dollar bet: “Over a ten-year period commencing on January 1, 2008, and ending on December 31, 2017, the S&P 500 will outperform a portfolio of funds of hedge funds, when performance is measured on a basis net of fees, costs and expenses.”

Buffet picked Vanguard’s S&P 500 Index FundTom Seides of Protégé Partners chose collection of hedge funds

Who Won? By How Much?

Buffet's $1m dollar bet: "Over a ten-year period commencing on January 1, 2008, and ending on December 31, 2017, the S&P 500 will outperform a portfolio of funds of hedge funds, when performance is measured on a basis net of fees, costs and expenses."

Buffet picked Vanguard's S&P 500 Index Fund with expense ratio of 0.04%

Tom Seides of Protégé Partners chose collection of hedge funds, ~2.5% annual fees.

Buffet won. Decisively.

Here's the final scorecard for the bet:

Year	Fund-of-Funds A	Fund-of-Funds B	Fund-of-Funds C	Fund-of-Funds D	Fund-of-Funds E	S&P Index Fund
2008	-16.5%	-22.3%	-21.3%	-29.3%	-30.1%	-37.0%
2009	11.3%	14.5%	21.4%	16.5%	16.8%	26.6%
2010	5.9%	6.8%	13.3%	4.9%	11.9%	15.1%
2011	-6.3%	-1.3%	5.9%	-6.3%	-2.8%	2.1%
2012	3.4%	9.6%	5.7%	6.2%	9.1%	16.0%
2013	10.5%	15.2%	8.8%	14.2%	14.4%	32.3%
2014	4.7%	4.0%	18.9%	0.7%	-2.1%	13.6%
2015	1.6%	2.5%	5.4%	1.4%	-5.0%	1.4%
2016	-3.2%	1.9%	-1.7%	2.5%	4.4%	11.9%
2017	12.2%	10.6%	15.6%	N/A	18.0%	21.8%
Final Gain	21.7%	42.3%	87.7%	2.8%	27.0%	125.8%
Average Annual Gain	2.0%	3.6%	6.5%	0.3%	2.4%	8.5%

Passive Investing & Fees

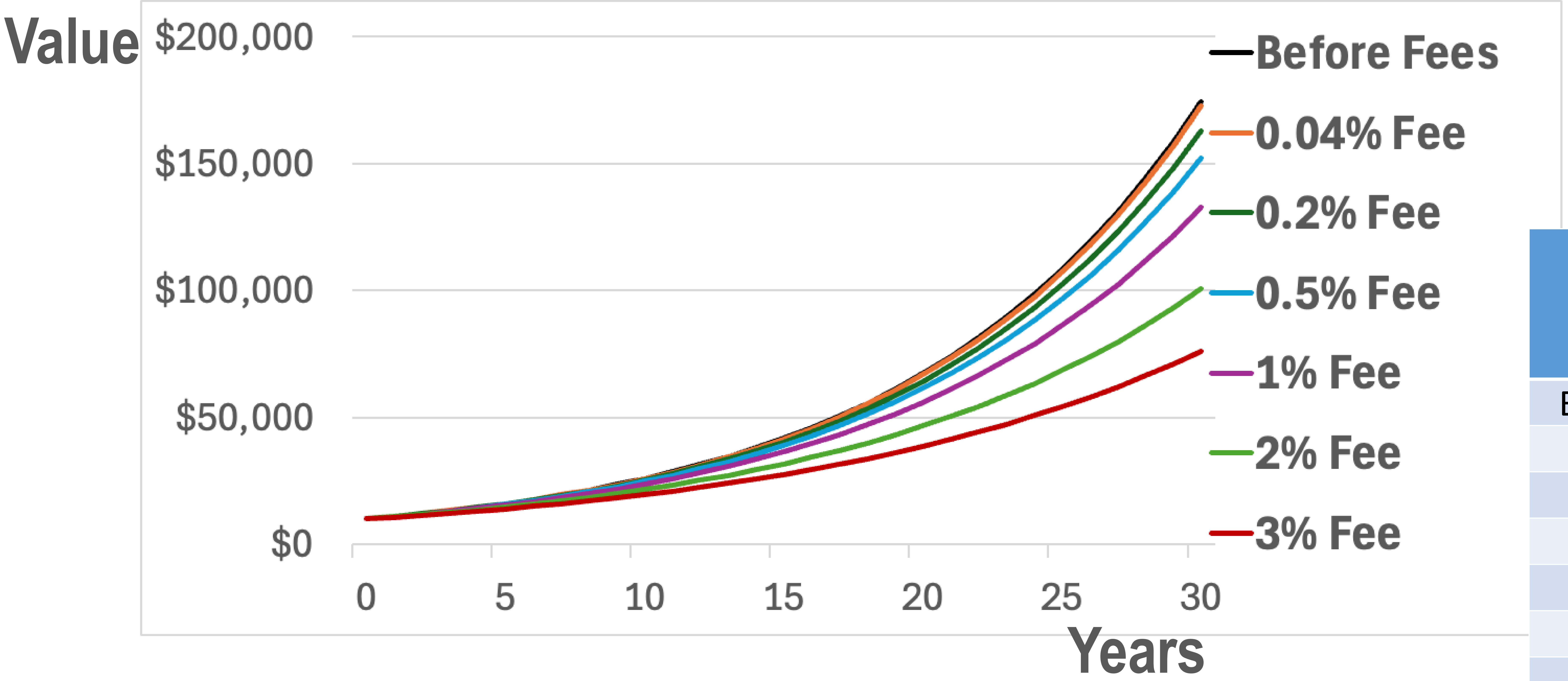
Investors should not be able to predictably outperform the market by implementing a tradeable strategy on a risk-adjusted basis.

Lots of smart people with better data than you are trying to find mispricing. That makes markets efficient so you can just free-ride on following index!

Imagine you invest $10,000 in the S&P500 Index. Expected returns are 10% a year. How much of a difference do fees make to your return over 30 years?

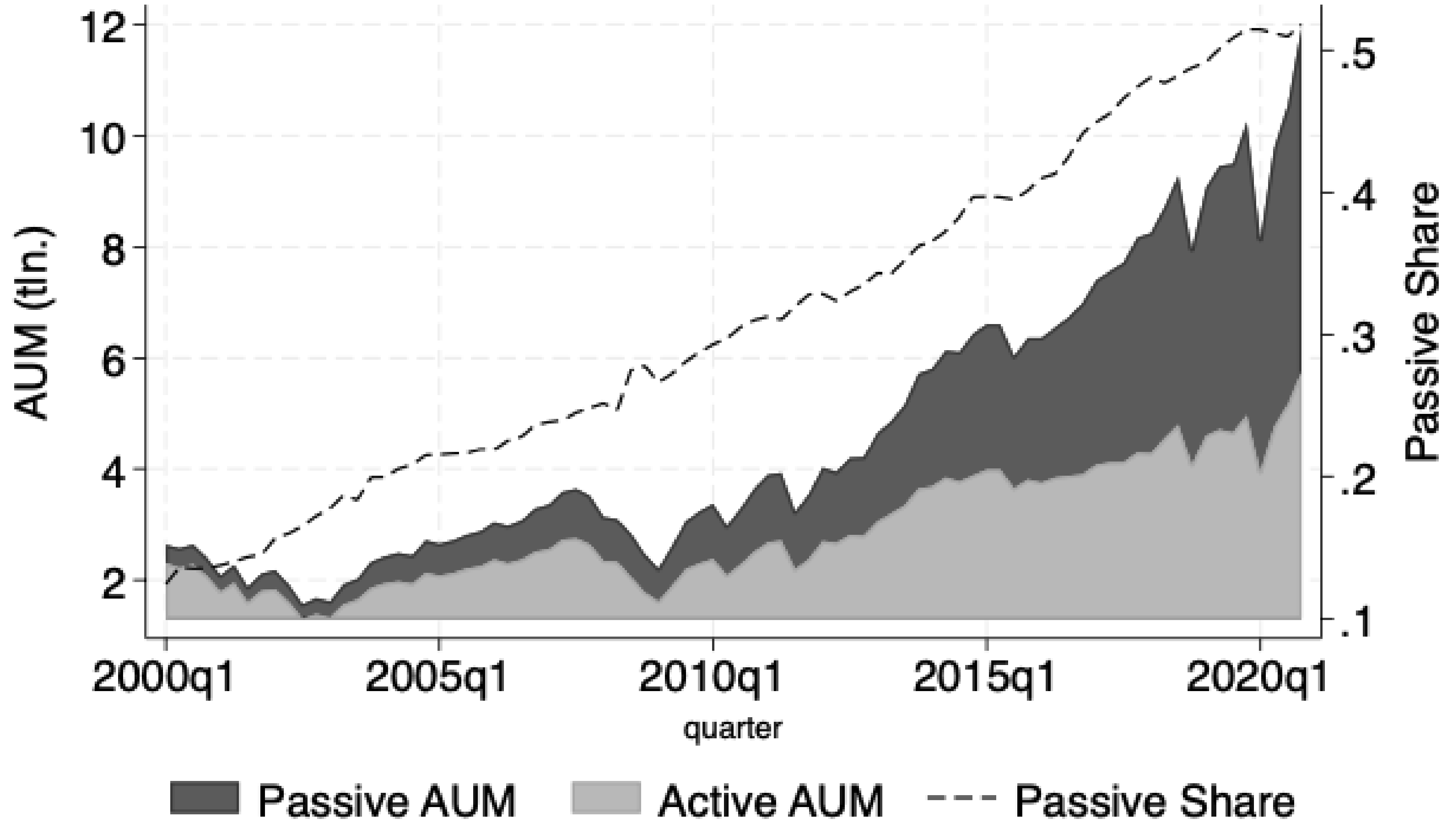
Passive Investing & Fees

Imagine you invest $10,000 in the S&P500 Index. Expected returns are 10% a year. How much of a difference do fees make to your return over 30 years?



	Value of $10,000 investment after 30 Years
Before Fees	$174,494
0.04% Fee	$172,600
0.2% Fee	$162,981
0.5% Fee	$152,203
1% Fee	$132,677
2% Fee	$100,627
3% Fee	$76,123

Growth in Passive Investing & Decline in Fees



Once again, Vanguard is lowering the cost of investing. Effective February 1, 2025, the firm reduced fees on 168 share classes across 87 funds. The fee reductions are expected to save investors more than $350 million this year alone.[1]

Why investment costs matter

Vanguard Founder John C. Bogle explained why investment costs matter this way:

In investing, realize that you get what you don't pay for. Whatever future returns the markets are generous enough to deliver, few investors will succeed in capturing 100% of those returns, simply because of the high costs of investing—all those commissions, management fees, investment expenses, yes, even taxes—so pare them to the bone.[2]

Efficient Markets?
Behavioral Finance

Behavioral Finance

- Seeking to explain financial behavior
- Found 'anomalies' where financial data diverged from financial theory's predictions.
- Understand data by combining economics with psychology.

- Huge psychology literature not covering today including…
    - Loss Aversion (losses hurt more than gains)
    - Biased Beliefs (incorrect beliefs about future given information potentially available)
    - Emotions (good vibes / regret)
    - Limited Attention (finite capacity to process information, focus on salient / available / recent)

EMH

“No Free Lunch” – You Can’t Beat the Market

-> Approximately true. Is hard to consistently beat the market.

“The Price is Right” – Asset Prices are Equal to Intrinsic Value

-> If not correct, then capital is not being correctly allocated across markets.

EMH

“No Free Lunch” – You Can’t Beat the Market

-> Approximately true. Is hard to consistently beat the market.

“The Price is Right” – Asset Prices are Equal to Intrinsic Value

-> If not correct, then capital is not being correctly allocated across markets.

“Limits to arbitrage” means that restrictions on investor behavior may prevent prices from correcting for some time

- Fundamental Risk: Risk of new information about an asset that you are trying to arbitrage (potentially with leverage)

- Noise Trader Risk: May need to close out position if mispricing worsens/continues for a long time.

- Costs: Trading costs, short-selling costs, costs of paying quants to detect, understand an exploit mispricing

January Effect (Keim, 1983)

- Stocks, especially small stocks, generate abnormal returns in January

D.B. Keim, Stock return seasonality and the size effect 21

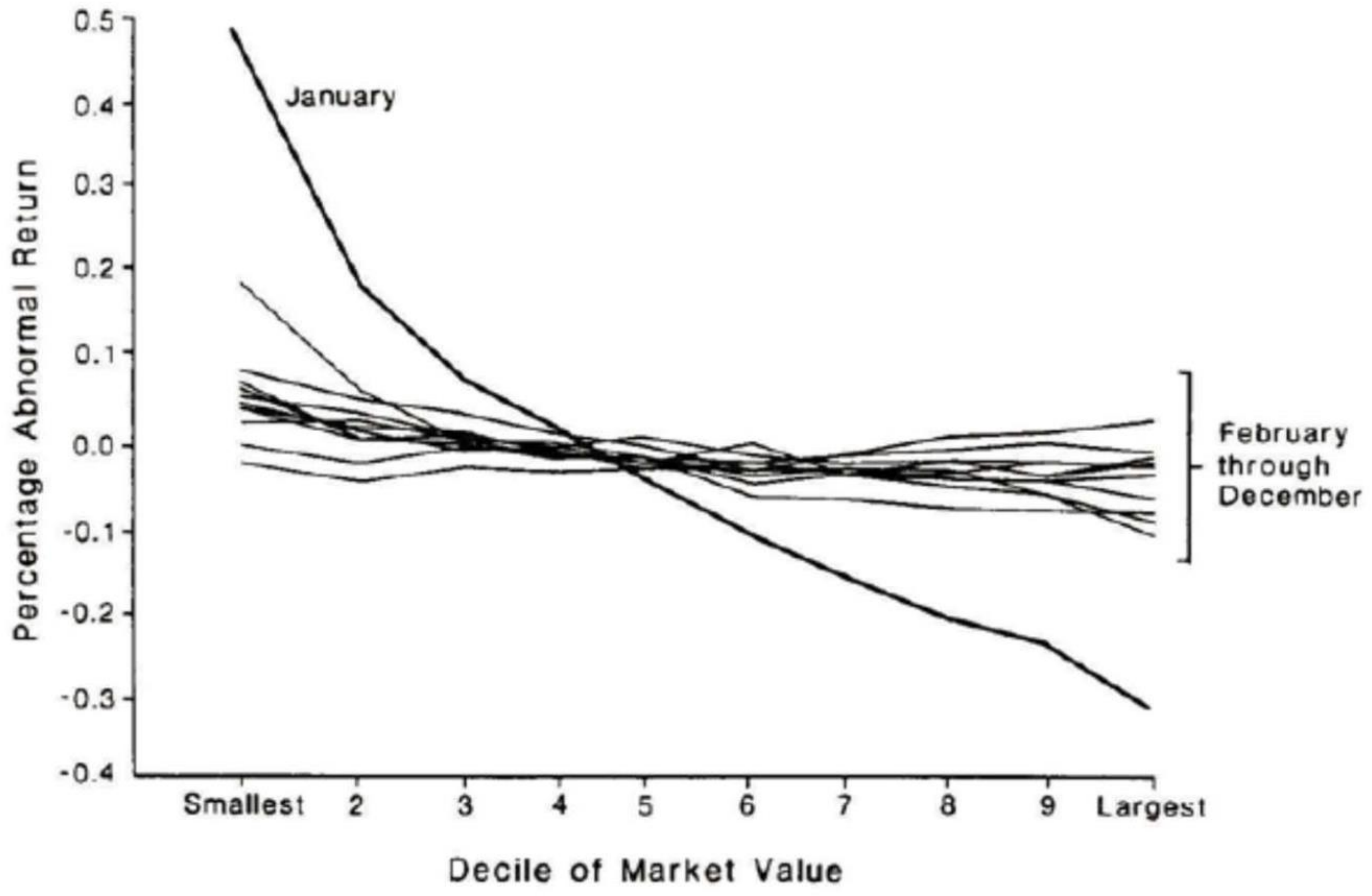


Fig. 2. The relation between average daily abnormal returns (in percent) and decile of market value *for each month* over the period 1963–1979. The ten market value portfolios (deciles) are constructed from firms on the NYSE and AMEX. Abnormal returns are provided by CRSP.

Jones Graduate School of Business

January Effect (Keim, 1983)

- Stocks, especially small stocks, generate abnormal returns in January
- People sell in December for tax losses, buy in January (investing year-end bonuses)
- Not just taxes or bonuses, hold in other countries without these features
- Also see weekend effects (Fridays good, Mondays bad), holidays (good before) etc

D.B. Keim, Stock return seasonality and the size effect 21

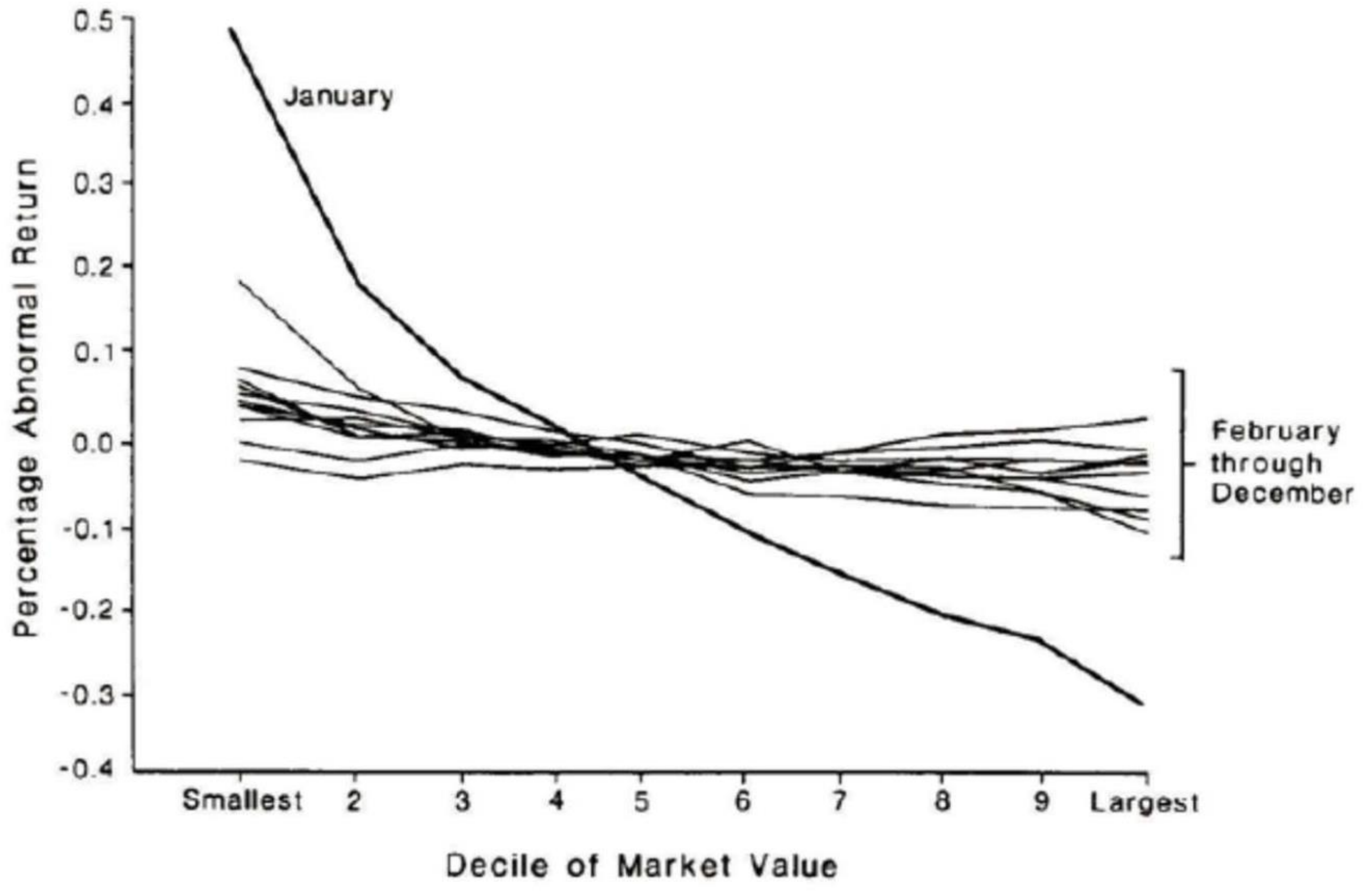


Fig. 2. The relation between average daily abnormal returns (in percent) and decile of market value *for each month* over the period 1963–1979. The ten market value portfolios (deciles) are constructed from firms on the NYSE and AMEX. Abnormal returns are provided by CRSP.

RICE | BUSINESS
Jones Graduate School of Business

Equity Premium Puzzle (Mehra & Prescott, 1985)

- Stocks outperform Treasury Bills by more than standard models can explain
- Implies implausibly high relative risk aversion (RRA).
  - If gamble (A) 50% to double your wealth, (B) 50% half your wealth.
  - Equity premium puzzle RRA implies you are willing to pay 49% of wealth to avoid gamble.
- If you believe the equity premium is a fair return for bearing the risks of stocks then choose assets based on you think you are more/less risk averse than marginal investo
- If you think that the equity premium is partially derived from other peoples' mistakes and fears, -> equities are very attractive investment!

Excess Volatility Puzzle (Shiller 1981; 2003; Campbell & Shiller, 1988)

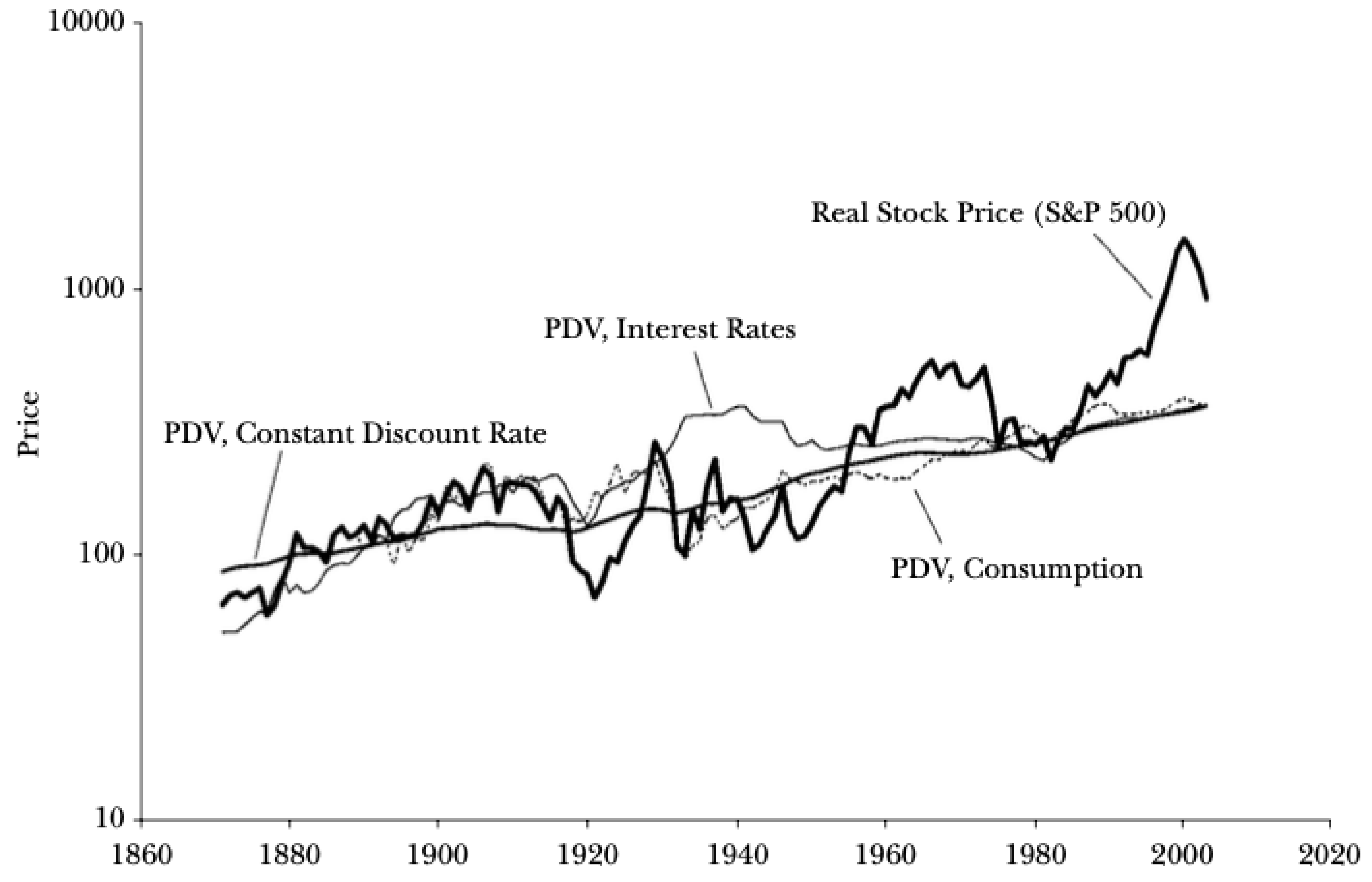
Stock prices are too volatile to be explained by fluctuations in expected dividends.

VAR(P) > VAR (PV of future dividends)

Figure 1

Real Stock Prices and Present Values of Subsequent Real Dividends

(*annual data*)

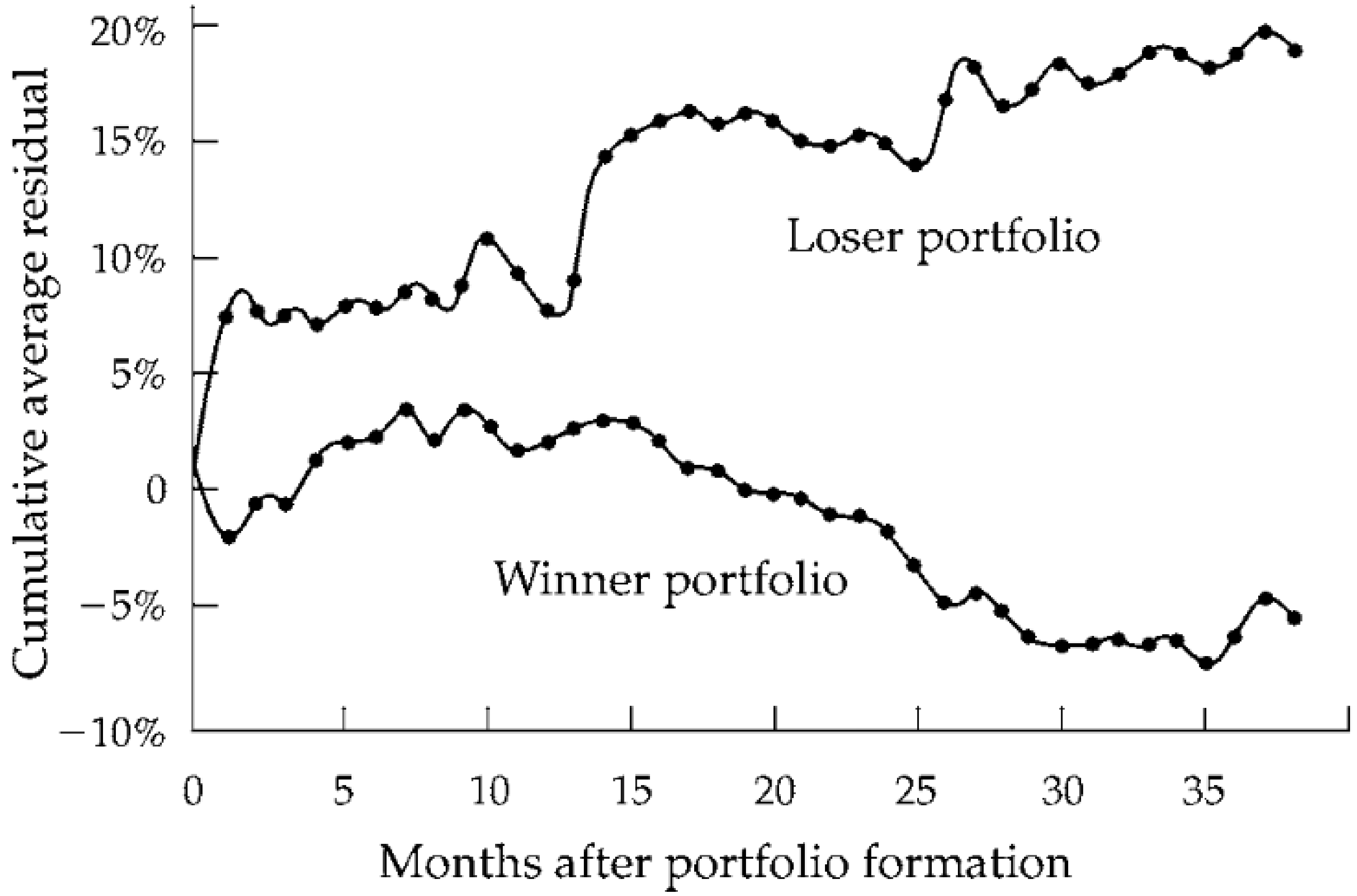


RICE | BUSINESS
Jones Graduate School of Business

Long-Term Reversals (DeBondt & Thaler, 1985)

Create portfolios of best (Winner) and worst (Loser) performing stocks over last 3 years. Compute 5-year returns on portfolio

Losers outperform, as market overreacts

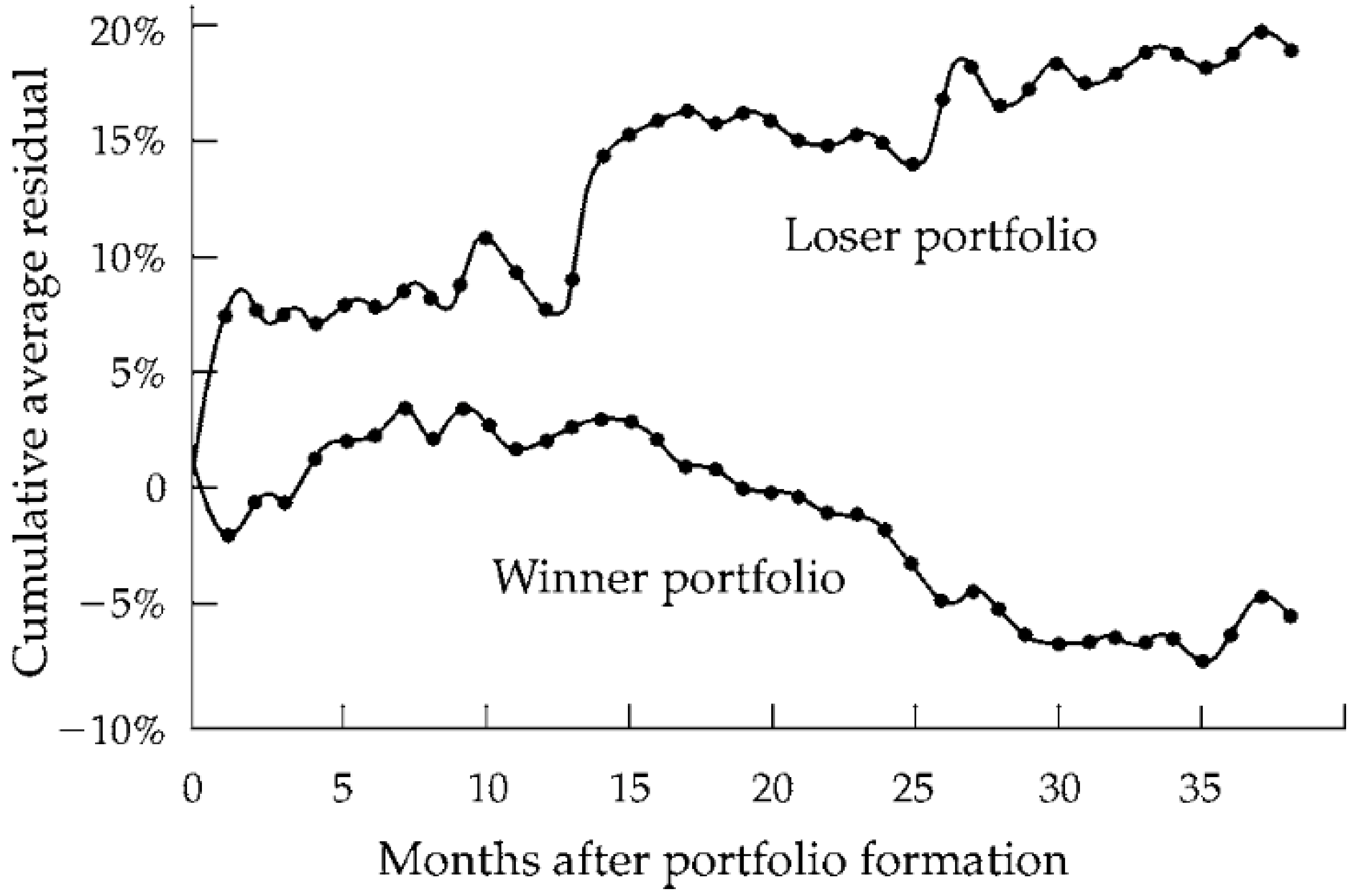


RICE | BUSINESS
Jones Graduate School of Business

Long-Term Reversals (DeBondt & Thaler, 1985) and Medium-Term **Momentum** (Jegadeesh & Titman, 93, 01, 23)

Create portfolios of best (Winner) and worst (Loser) performing stocks over last 3 years. Compute 5-year returns on portfolio

Losers outperform, as market overreacts



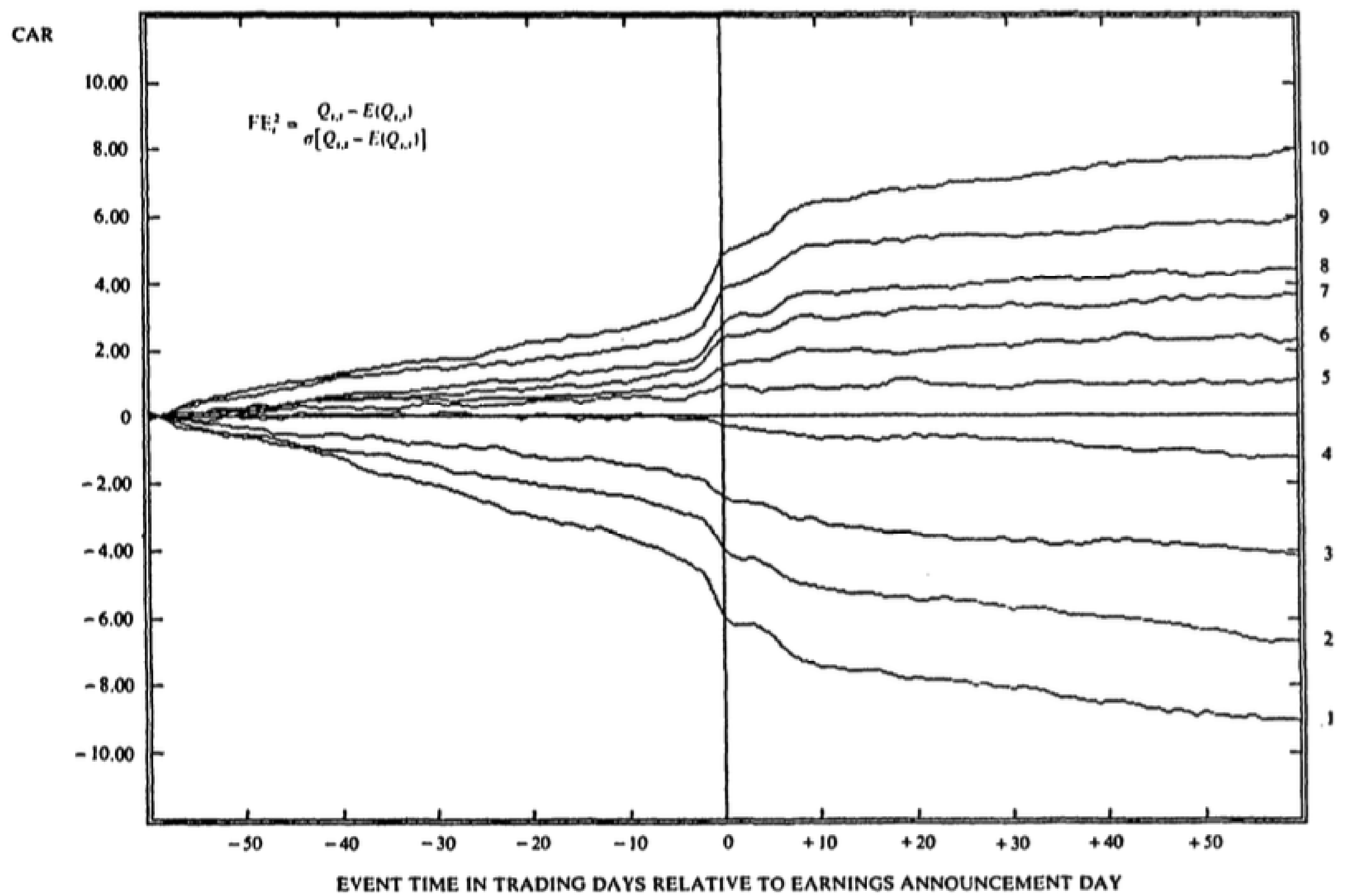
Excess returns in last 3-12 months predict excess returns in the next 3-12 months

Still holds using data to 2020

Country	Alpha	Beta
USA	0.74	-0.64
Australia	2.19	-0.02
Canada	1.02	-0.40
China	-0.06	-0.04
France	1.39	-0.43
Germany	2.21	-0.39
Italy	2.03	-0.45
Japan		
UK	0.74	-0.64

Post-Earnings Announcement Drift (Bernard & Thomas, 1989)

- Efficient Markets predict immediate price adjustment to new information, no drift.
- Rank firms by their earnings news (10 highest unexpected, 1 lowest)
- Show their cumulative abnormal return
- (An example of ‘**event study**’ approach - many others testing other announcements)



Good News Firms’ Returns Drift Up 🙂

Bad News Firms’ Returns Drift Down 🙁

Before Shell PLC, it had stocks traded in 2 countries:
- Royal Dutch (NL) & Shell (GB) - Froot & Dabora, 1999

These two companies pooled their cashflows.

Before Shell PLC, it had stocks traded in 2 countries: - Royal Dutch (NL) & Shell (GB) - Froot & Dabora, 1999

These two companies pooled their cashflows.

But their stock prices differed!

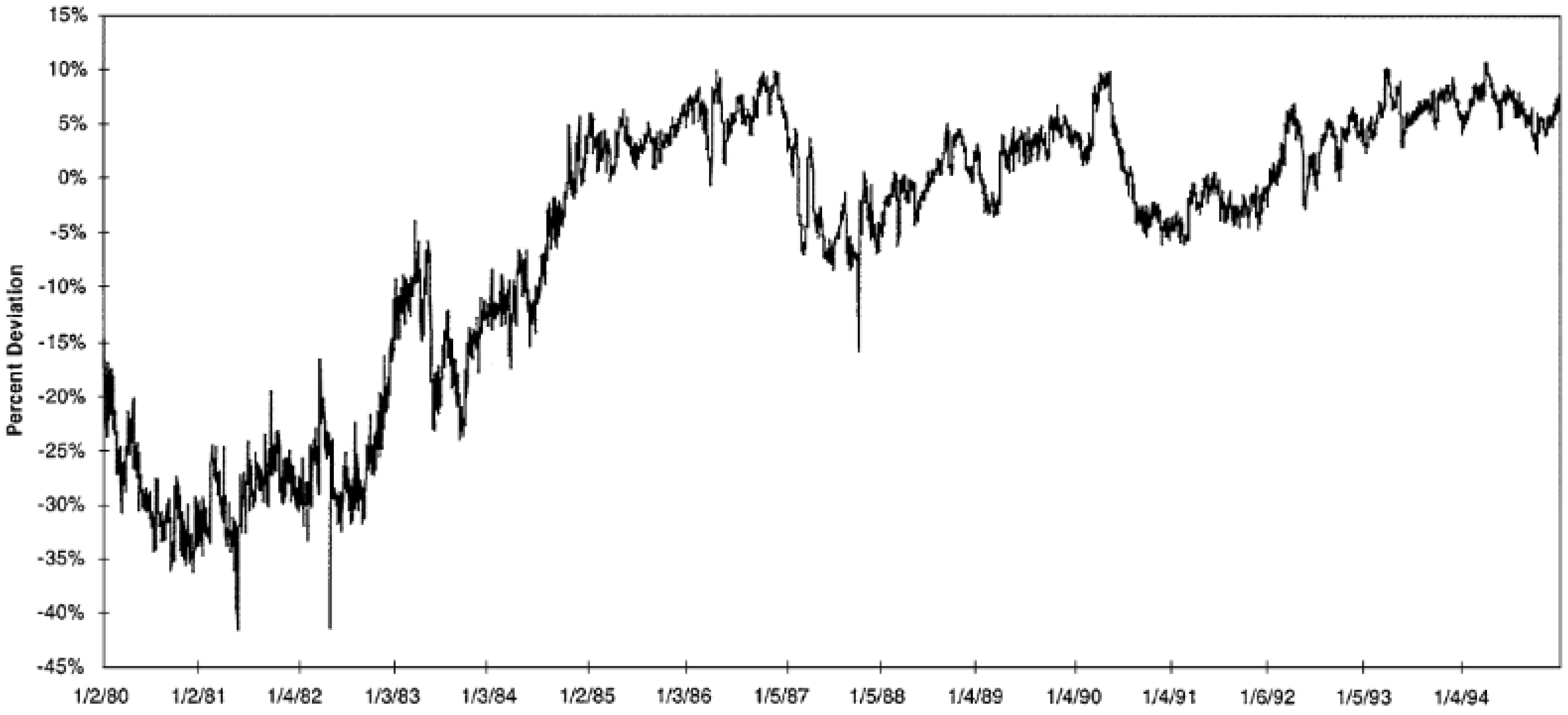


Fig. 1. Log deviations from Royal Dutch/Shell parity. Note: This figure shows on a percentage basis the deviations from theoretical parity of Royal Dutch and Shell shares and ADRs traded on the NYSE. Data are from the Center for Research in Security Pricing (CRSP).

"Massively Confused Investors Making Conspicuously Ignorant Choices (MCI–MCIC)" – Rashes 2002

MCIC: stock market ticker for large telecoms company (~$20bn in 1997)

MCI: stock market ticker for Massmutual Corporate Investors (~$200m net assets)

- Invests in long-term corporate debt. Has does not (and has never) held securities in MCIC or other major telecoms stocks.

Correlations in Stock Price Movements (Daily, 1994 - 1997):

- MCIC vs. AT&T:
- MCI vs. NYSE:
- MCIC vs. MCI:

"Massively Confused Investors Making Conspicuously Ignorant Choices (MCI–MCIC)" – Rashes 2002

MCIC: stock market ticker for large telecoms company (~$20bn in 1997)

MCI: stock market ticker for Massmutual Corporate Investors (~$200m net assets)

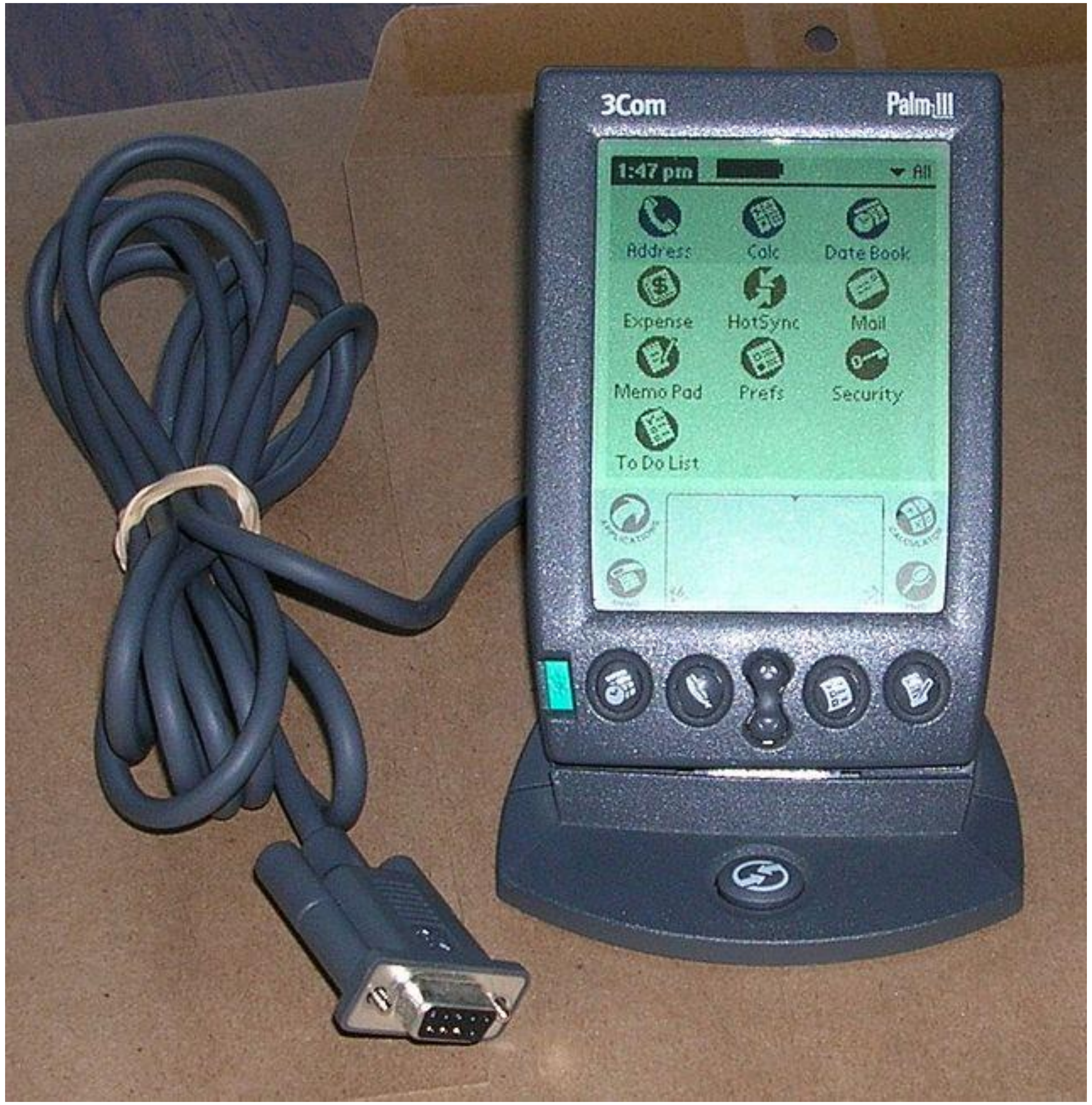
- Invests in long-term corporate debt. Has does not (and has never) held securities in MCIC or other major telecoms stocks.

Correlations in Stock Price Movements (Daily, 1994 - 1997):

- **MCIC vs. AT&T: 0.16**
- **MCI vs. NYSE: 0.12**
- **MCIC vs. MCI: 0.56!!!**

Can't be explained by a common 'fundamental factor'

Do You Know What This Is?



RICE | BUSINESS
Jones Graduate School of Business

“Can the Market Add and Subtract? Mispricing in Tech Stock Carve-Outs” – Lamont & Thaler 2003

Shareholders of Company A are each to receive 1.525 share of company B.

“Law of One Price” means identical assets *should* have identical prices.

So, if company B shares trade at $95.06 per share then

the price of company A should be the value of company A and also the value of company B’s shares that shareholders will receive: 1.525 x $95.06 = $145.

But what if share price of company A is <$145?!

This happened!

Company A (3Com), Company B (Palm). 3Com’s value was $81.81, implying the rest of its business was -$22bn. Gap persisted for months.

Shorting costs explain why mispricing occurs. Irrational investors explain why buying at wrong price!

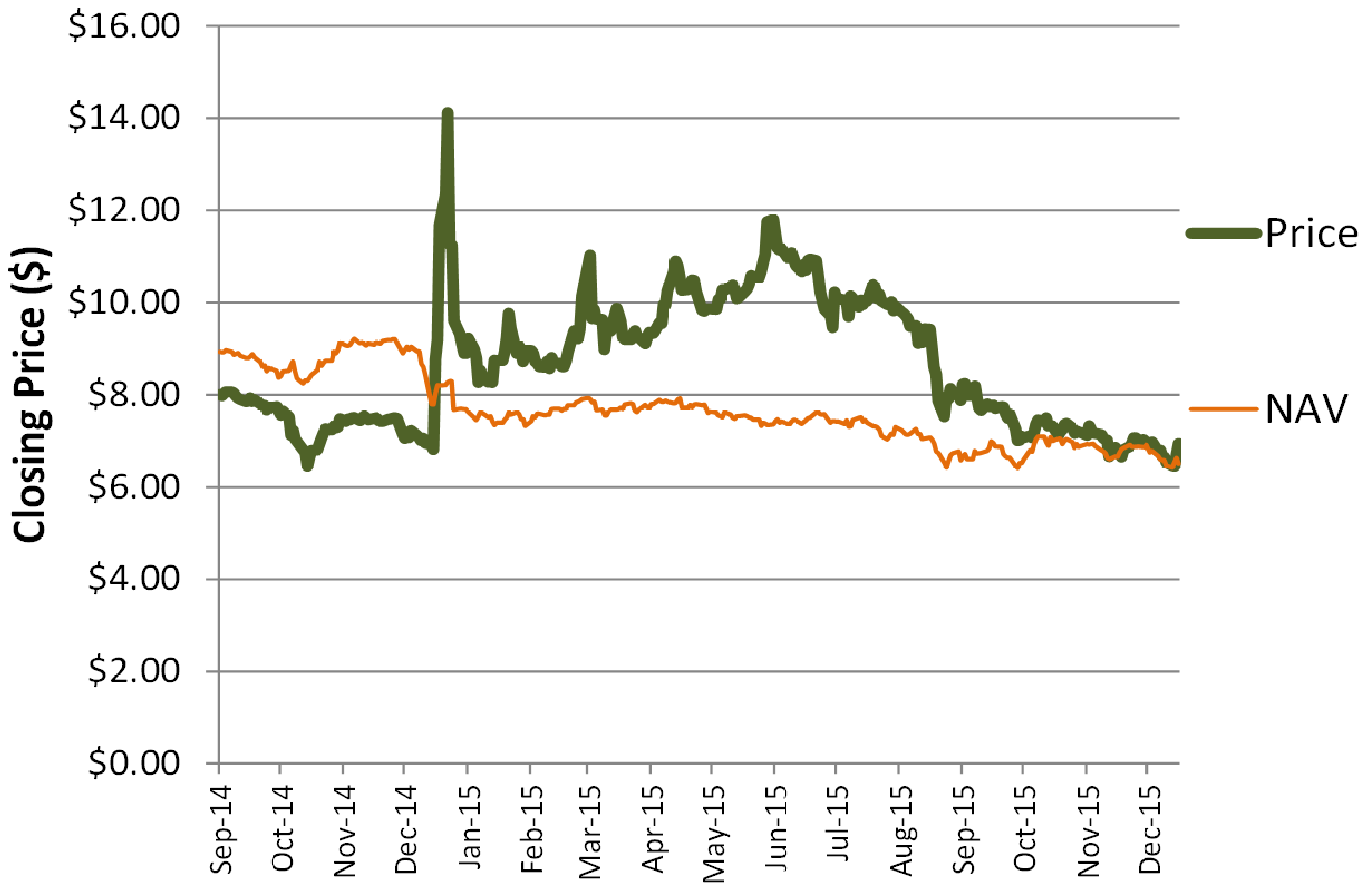
https://www.youtube.com/watch?v=bM9bYOBuKF4&t=549s

CUBA (Thaler)

Cuba Fund: A small closed-end fund, the Herzfeld Caribbean Basin Fund, has 69% of its holdings in US stocks with the rest in foreign stocks, chiefly Mexican.

Founded in 1994, it gave itself the ticker “CUBA” despite the fact that it owns no Cuban securities nor was it legal for any US company to do business in Cuba.

On December 17th 2014, Obama announced he was going to lift several restrictions against Cuba.



RICE | BUSINESS
Jones Graduate School of Business

Retail Investors

“Dumb Money” – Frazzini & Lamont (2008)

Follow retail investors’ mutual funds flows:

- “Individual investors have a striking ability to do the wrong thing.
- They send their money to mutual funds which own stocks that do poorly.
- Individual investors are **dumb money**
- Can use their mutual fund reallocation decisions to predict future stock returns.
- Individual investor sentiment causes some stocks to be misvalued relative to other stocks, and firms exploit this mispricing.”

“Just How Much Do Individual Investors Lose by Trading?” Barber, Lee, Liu & Odean (2009)

- Individual investor losses are 3.8 percentage points per year. 2.2% of Taiwan’s GDP.
- Losses due to overtrading
- Institutions 1.5 percentage point gains per year.

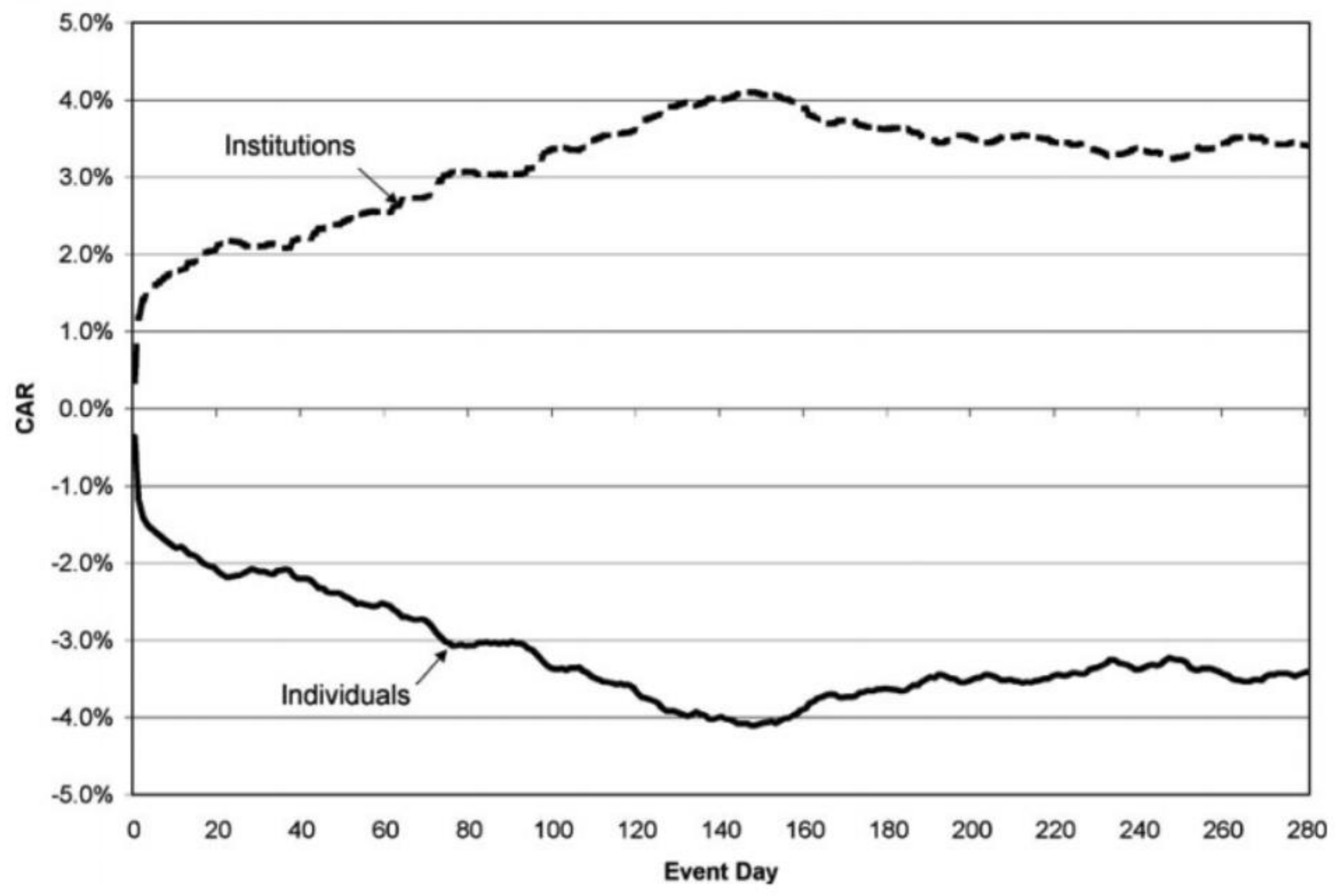


Figure 1
Cumulative (market-adjusted) abnormal returns (CARs) in event time for stocks bought less stocks sold by institutions and individuals
Panel A: CARs are weighted by aggregate value of stocks bought and stocks sold. Panel B: CARs are weighted by net value of stocks bought and sold.

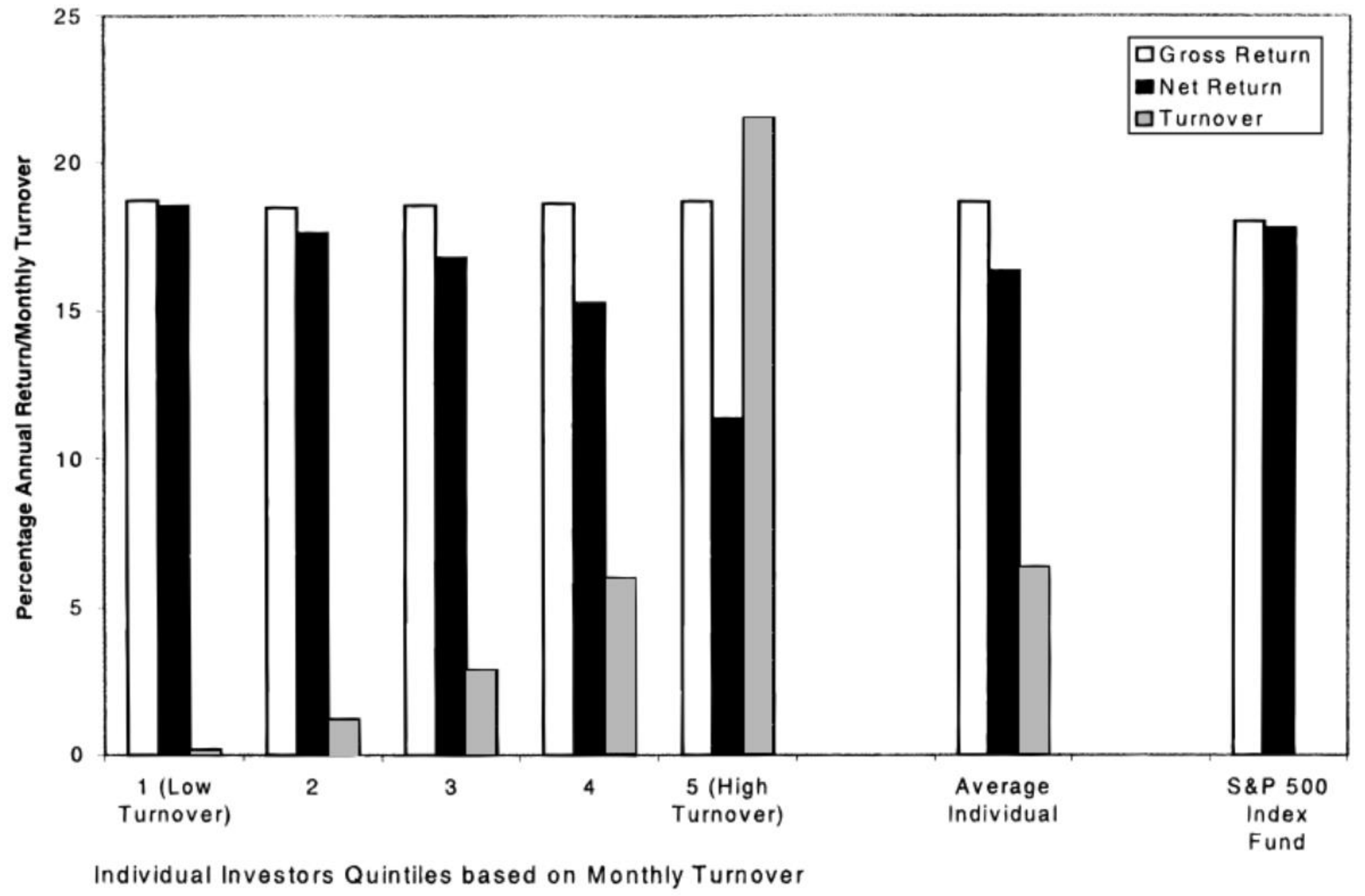
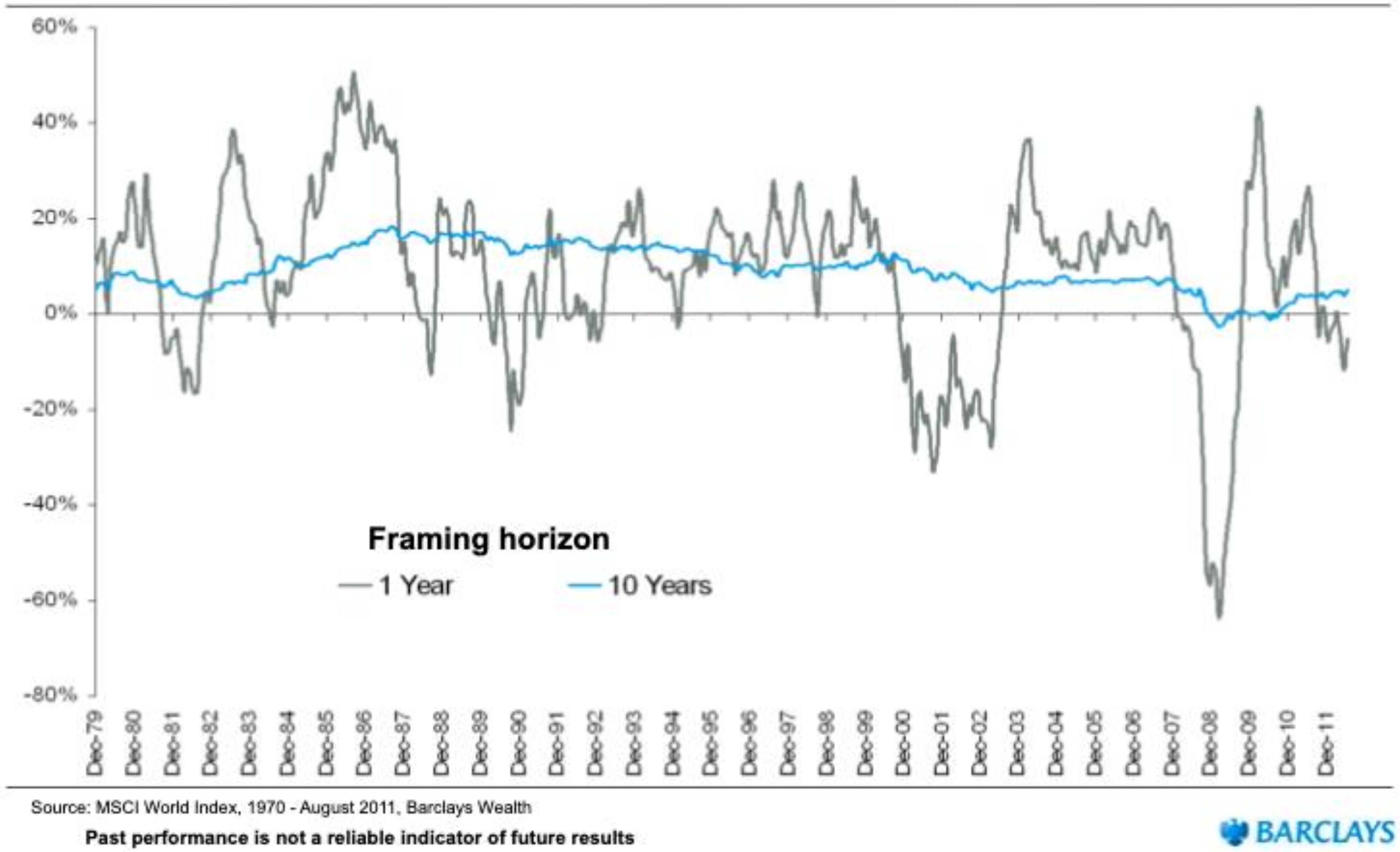
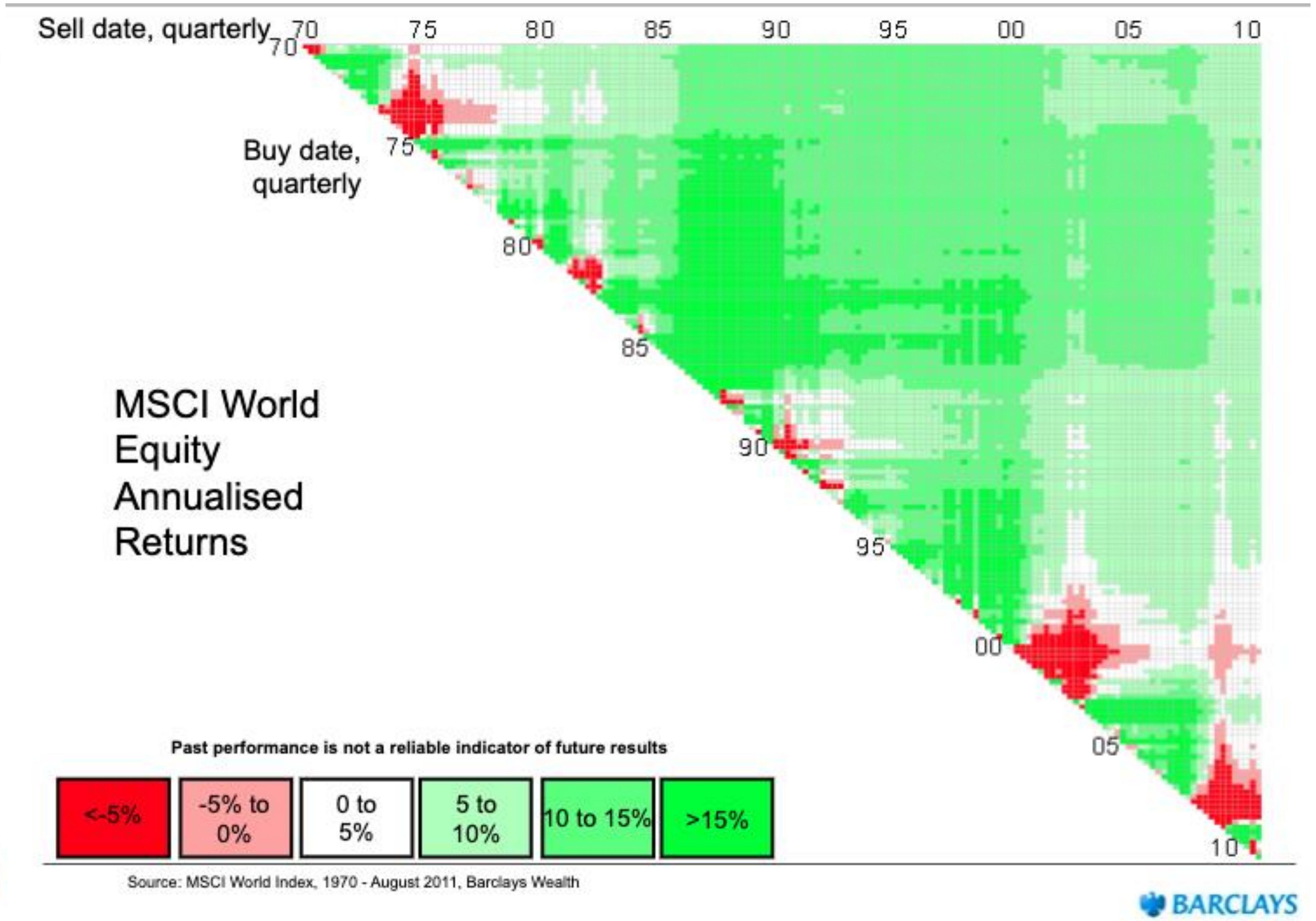


Figure 1. Monthly turnover and annual performance of individual investors. The white bar (black bar) represents the gross (net) annualized geometric mean return for February 1991 through January 1997 for individual investor quintiles based on monthly turnover, the average individual investor, and the S&P 500. The net return on the S&P 500 Index Fund is that earned by the Vanguard Index 500. The gray bar represents the monthly turnover.

Barber & Odean, 2000

Framing Matters





RICE | BUSINESS
Jones Graduate School of Business

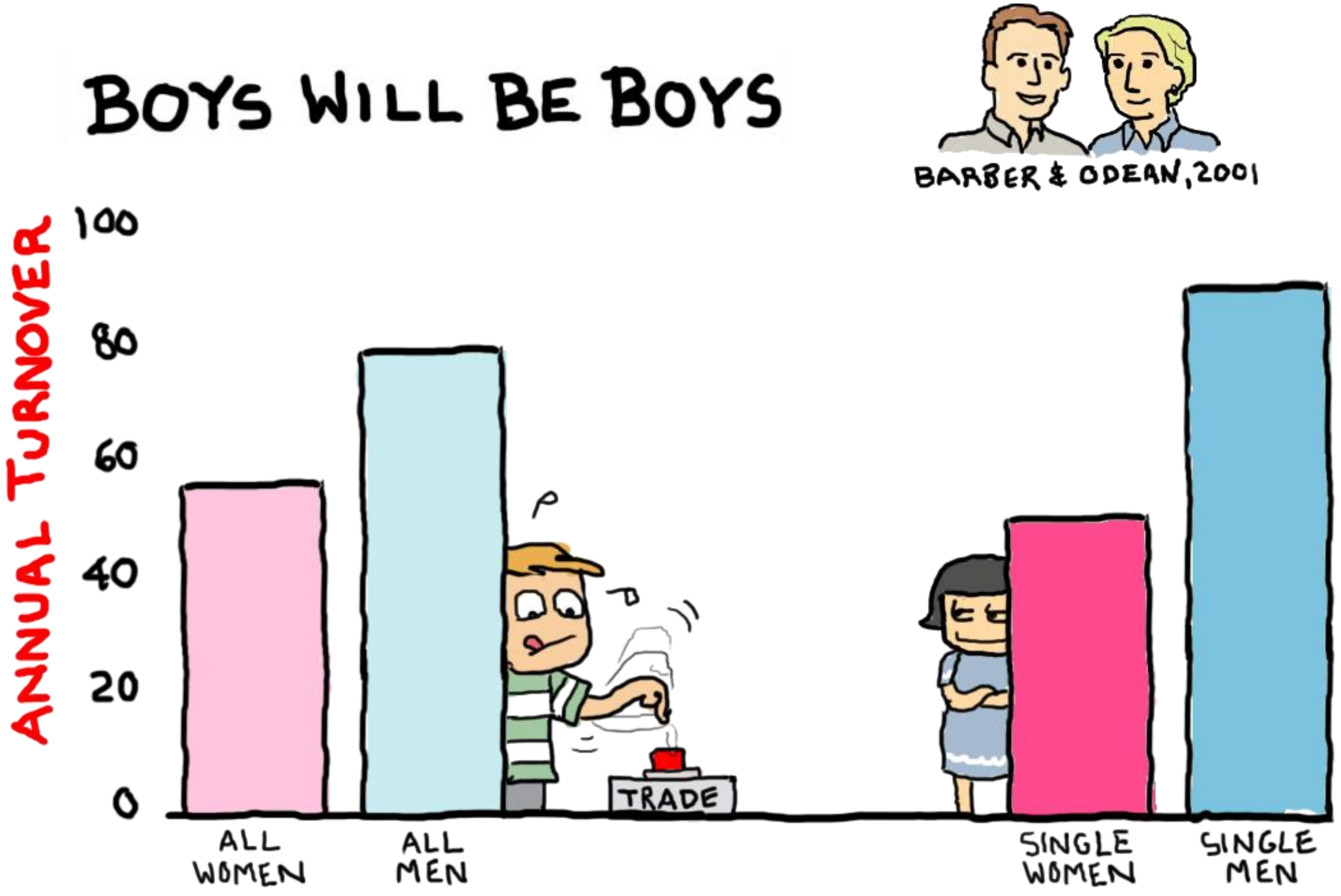
(Davies 2013)

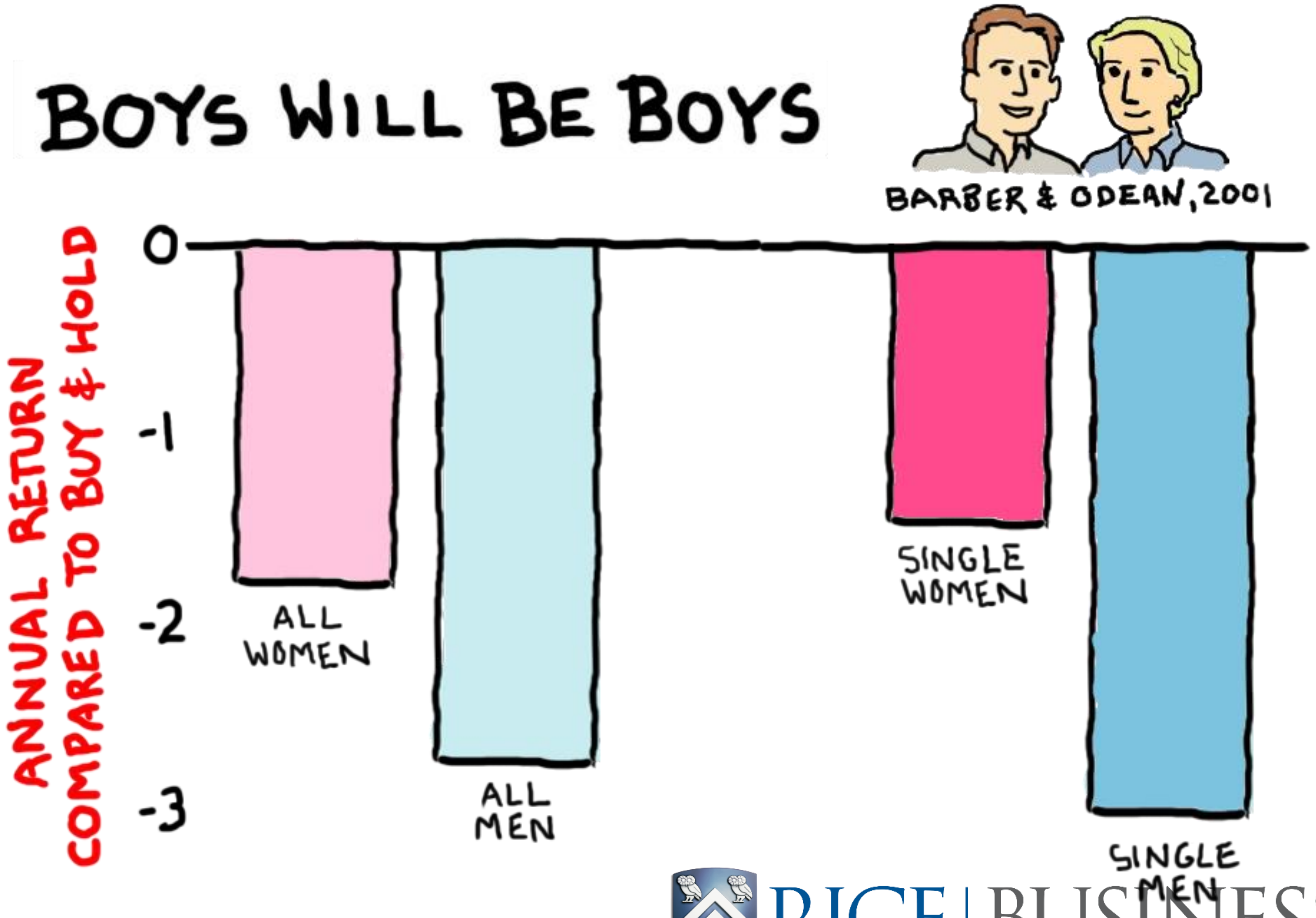
Home Bias & Naïve Diversification

- Portfolio Theory taught us benefits of diversification
- "Hedge": Can insure against risk (of losing income) by avoiding stocks correlated (with your income).
- Retail investors have a **"Home Bias"** investing in domestic stocks (94% US), under-diversification in international stocks (and also historically regions)
- Retail investors use simple rules-of-thumb to **naïvely diversify** their investments.
- Suppose you have 10 funds, investors put 10% in each.
  - If 1 of those 10 funds is bonds, 10% bonds / 90% equity
  - If 2 of those 10 funds is bonds, 20% bonds / 80% equity
  - If 5 of those 10 funds is bonds, 50% bonds / 50% equity

“Boys Will Be Boys” Barber & ODean (2000,2001)

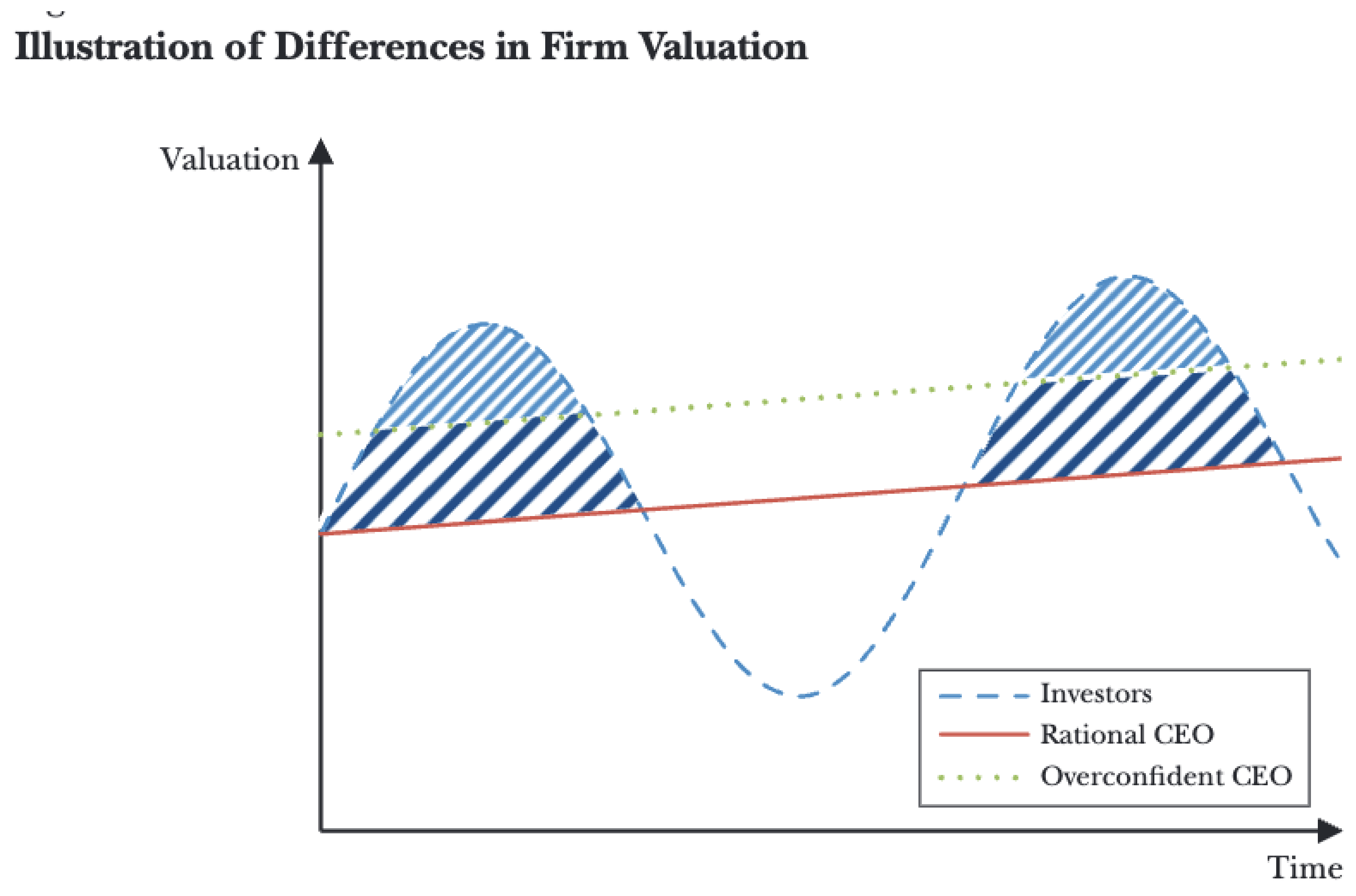
- Robust psychological finding that men more likely to be overconfident than women
- Men trade more excessively than women (larger gaps single men vs single women)
- More trading leads to lower returns





RICE | BUSINESS
Jones Graduate School of Business

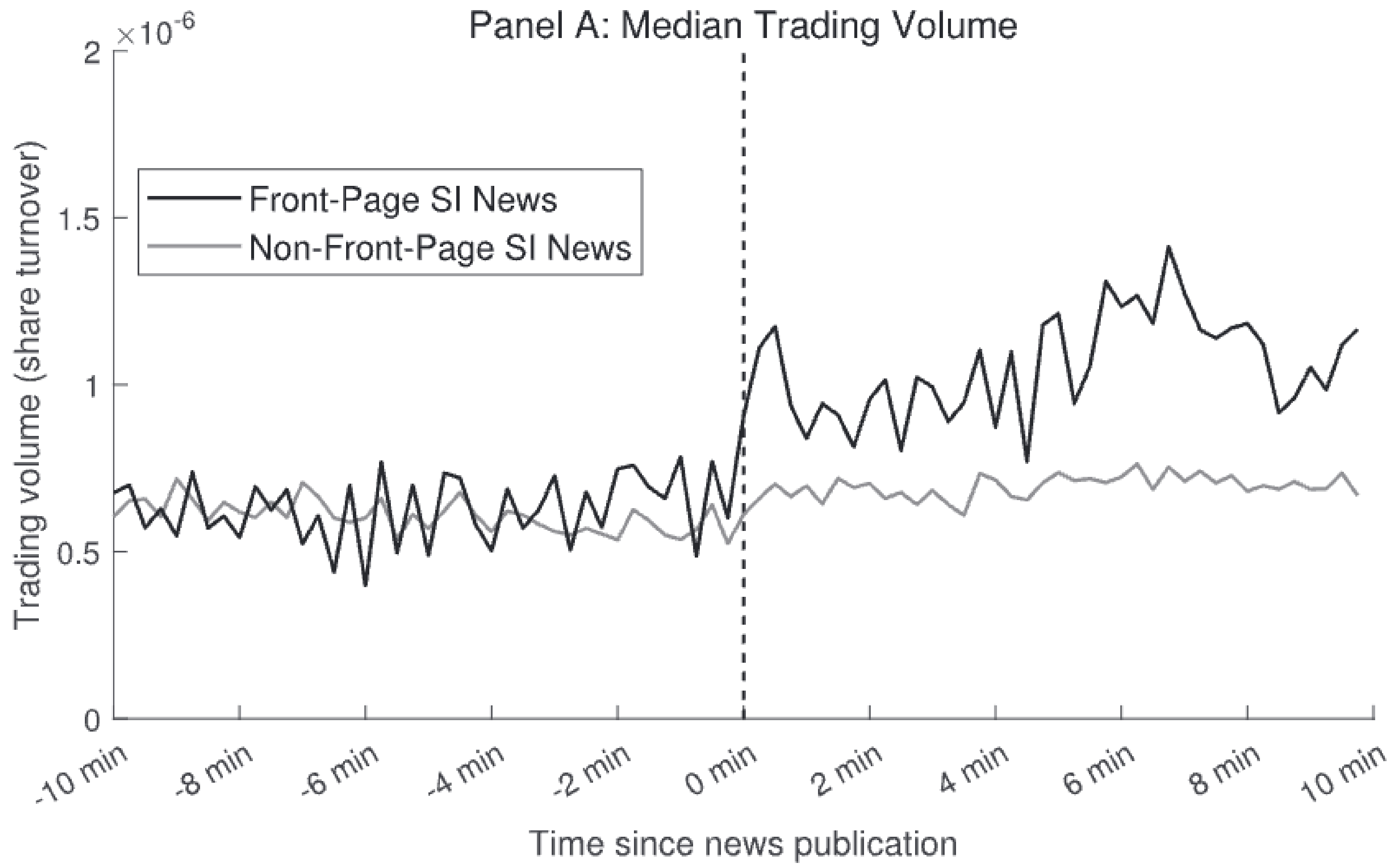
Overconfident CEOs (Malmendier & Tate, 2005)

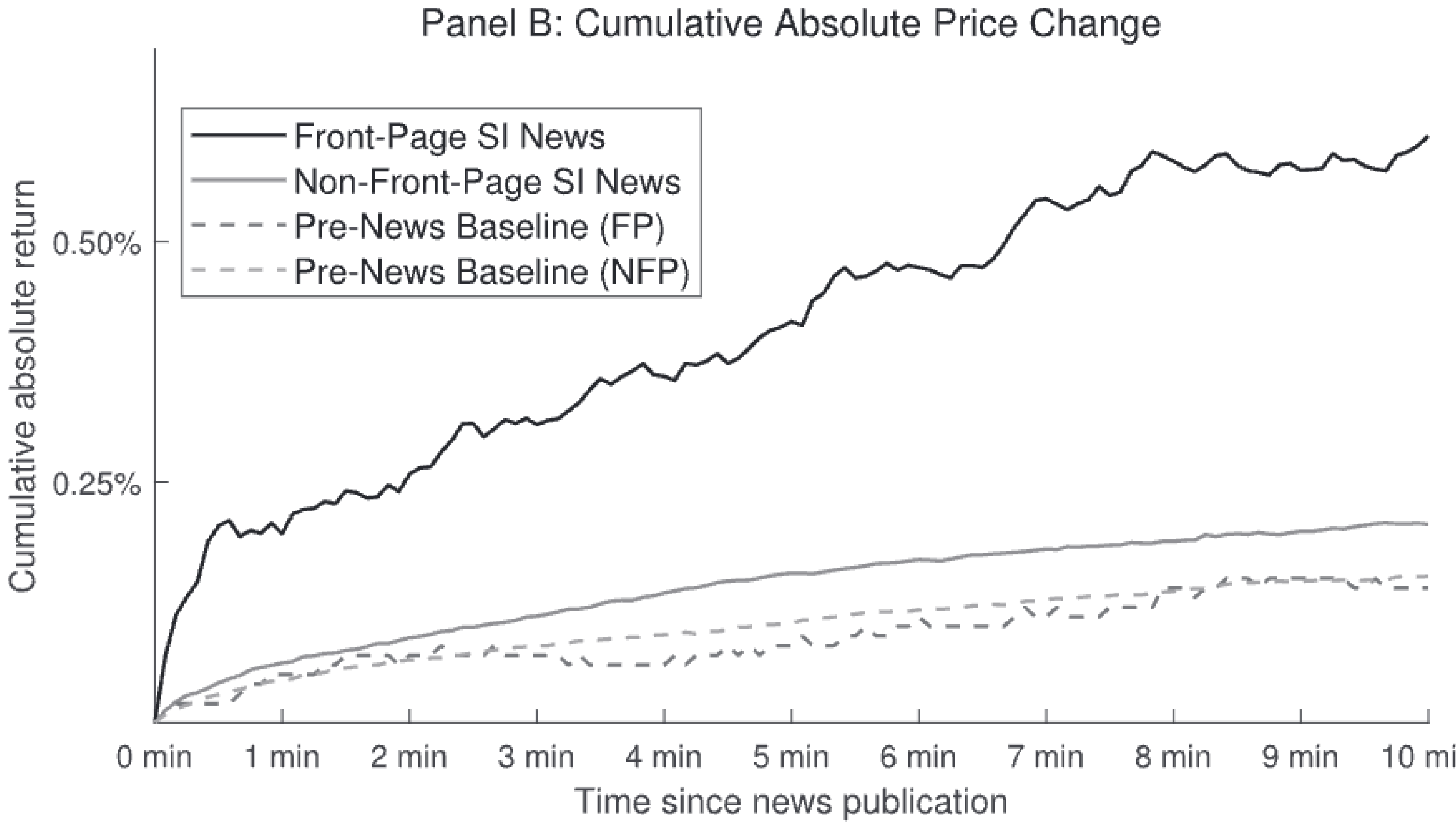


- Measure overconfidence if do not reduce personal exposure to company-specific risk.
- Overconfident CEOs overestimate the returns to *their* investment projects and view external funds as unduly costly.
- They overinvest when internal funds, under-invest if need external funds
- Real effects on corporate investments
- Companies are made up of (and led by) humans!

Effect of News Positioning on Financial markets (Fedyk, 2023)

- Study how positioning of news articles on Bloomberg terminal affects trading.
- Bloomberg terminals used by sophisticated traders.
- Front-page articles get 240% higher trading volume & 176% higher excess returns in first 10 minutes, than equally important non-front-page news articles.
- Takes 2+ days for prices to fully reflect information.





Takeaways

Takeaways

- Efficient Market Hypothesis gives us a theory to benchmark
- Generally, EMH is a pretty good model of the world
- It is hard to make excess profits repeatedly.
- Wisdom of crowds (not just stock markets, see other prediction markets)
- BUT prices are not always right – there are some limits to arbitrage
- Retail investors often make poor investment decisions
- Psychology can help understand investor behavior

  - Not just uninformed retail investors, "informed" CEOs can also make poor decisions
- Real finance firms' developed based on ideas in this lecture (DFA, AQR)